RV College of Engineering®



Industrial Engineering & Management

Bachelor of Engineering (B.E)

Scheme And Syllabus Of VII & VIII Semester
(2022 Scheme)

B.E. Programs : AI, AS, BT, CD, CH, CS, CV, CY, EC, EE, EI, ET, IM, IS, ME.
M. Tech (13) MCA, M.Sc. (Engg.)
Ph.D. Programs : All Departments are recognized as Research Centres by VTU Except AI & AS

2025

RV College of Engineering®

SINCE 1963

NAAC GRADE A+ ACCREDITED INSTITUTION

SHAPING VISIONARIES, CRAFTING INNOVATORS



An autonomous institution affiliated with VTU, Belagavi, and approved by AICTE and UGC, New Delhi

- **RV SATELLITE-1 (RVSAT - 1)** India's First Microbiological Payload, launched by RVCE students in December 2024
- **NCC** - Best Institution Award of Karnataka and Goa Directorate (2019, 2022, 2023 & 2025)

₹50 Cr.

MoU with **TATA Technologies** to establish Centre for Invention, Innovation, Incubation & Training

Placements

₹92 LPA	250+	92%	₹10 LPA
Highest Package	Recruitment Partners	Placement Rate	Median Salary

Rankings

99th	1501+	601+	NBA	IIRF 2025	NPTEL
NIRF Ranking 2024 in Engineering Category	Times Higher Education World University Rankings-2025	Times Higher Education Asia University Rankings-2025	UG and PG programs accredited multiple times	Engineering ranking India National - 01 State - 01 Zone - 01	Local Chapter Ranking 2025 National – 09 State – 01

Research and Centres

37 Skill Labs

General Skills, Domain Skills, Interdisciplinary Skills

- Bosch Rexroth
- Mercedes Benz
- Morris Garage
- Boston AI System
- Toyota Kirloskar
- Vision Astra EV Academy

Outcome Based Education (OBE) emphasizing on experiential learning, research-based learning and skill-based learning

20	8
Centres of Excellence	Centres of Competence

80	65	46
Patents filed	Patents published	Patents granted

32	₹30 Cr.
Innovative & Cultural activity teams	Research and projects consultancy (last 5 years)

16 Research Centres in niche areas

- Autonomous Vehicles
- Nanomaterials and Devices
- Hydrogen and Green Technology
- Extended Reality
- Healthcare
- E-mobility
- IoT and Sensors
- Smart Antenna System
- Computational Genomics
- Quantum Computing

Programs and Faculty

- **13 UG** Programs; **1420 (2025)** annual sanctioned intake
- **13 PG** Programs **390 (2025)** annual sanctioned intake
- **Ph.D.** Programs in all departments
- **500+** Faculty Members and Staff

CET / KEA Code: **E005**
COMED-K Code: **E095**
PGCET Code (MTech): **T857**
PGCET Code (MCA): **C463**

State-of-the-Art infrastructure

- **Innovation Center**
- **Gymnatorium**
- **Hostels**
- **Labs & Workshops**

Alumni

36,000 +	100+
Alumni across the globe	RVCE alumni in civil service

RVCE Reconnect - USA edition, our first International Alumni Meet

NUMBER OF ATTENDEES

650	120
San Jose	Seattle

Mr. Suresh Katta, RVCE alumnus & Chairman Emeritus of Saama Group, USA and **Mr. Anil Kumble,** RVCE alumnus & former Indian cricketer, were the Guests of Honour at the alumni meets held in San Jose & Seattle, USA.

Collaborations with International universities

125 MoUs with Industry, Institutions & Research Establishments

thws Technical University of Applied Sciences Würzburg-Schweinfurt · Fachhochschule Dortmund University of Applied Sciences and Arts · Technische Hochschule Rosenheim · HTWD Hochschule für Technik und Wirtschaft Dresden University of Applied Sciences · The University of Iowa · FIU Florida International University · Saint Peter's University · Kennesaw State University · RWTH Aachen University

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post, Bengaluru - 560059, Karnataka, India | +91-80-68188110 | www.rvce.edu.in

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Go, change the world®

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Industrial Engineering & Management

Bachelor of Engineering (B.E)

Scheme And Syllabus Of VII & VIII Semester
(2022 Scheme)

B.E. Programs : AI, AS, BT, CD, CH, CS, CV, CY, EC, EE, EI, ET, IM, IS, ME.
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2025

INDUSTRIAL ENGINEERING & MANAGEMENT

DEPARTMENT VISION

Imparting innovation and value-based education in Industrial Engineering and Management for steering organizations to global standards with an emphasis on sustainable and inclusive development.

DEPARTMENT MISSION

1. To impart scientific knowledge, engineering and managerial skills for driving organizations to global excellence.
2. To promote a culture of training, consultancy, research and entrepreneurship interventions among the students.
3. To institute collaborative academic and research exchange programs with national and globally renowned academia, industries and other organizations.
4. To establish and nurture centers of excellence in the niche areas of Industrial and Systems Engineering.

PROGRAM EDUCATIONAL OBJECTIVES

PEO1	Conceive, design, implement and operate integrated systems, focus on appropriate measures of performance at strategic, tactical and operational levels.
PEO2	Develop competency to adapt to changing roles for achieving organizational excellence.
PEO3	Design and develop sustainable technologies and solutions for betterment of society.
PEO4	Pursue entrepreneurial venture with a focus on creativity and innovation for developing newer products, processes and systems.

PROGRAM SPECIFIC OUTCOMES

PSO1	Design, develop, implement and improve integrated systems that include people, Materials, information, equipment and energy.
PSO2	Apply statistical and simulation tools, optimization and meta heuristics techniques for analysis of various systems leading to better decision making.
PSO3	Demonstrate the engineering relationships between the management tasks of planning, Organization, leadership, control, and the human element in various sectors of economy.

LEAD SOCIETY

Institute of Industrial Engineers (IIE)

Sl. No.	Abbreviation	Meaning
1.	VTU	Visvesvaraya Technological University
2.	BS	Basic Sciences
3.	CIE	Continuous Internal Evaluation
4.	SEE	Semester End Examination
5.	PE	Professional Core Elective
6.	IE	Institutional Elective
7.	HSS	Humanities and Social Sciences
8.	PM	Physics
9.	CM	Chemistry
10.	MA	Mathematics
11.	AS	Aerospace Engineering
12.	AI&ML	Artificial Intelligence & Machine Learning
13.	BT	Biotechnology
14.	CH	Chemical Engineering
15.	CS	Computer Science & Engineering
16.	CV	Civil Engineering
17.	EC	Electronics & Communication Engineering
18.	EE	Electrical & Electronics Engineering
19.	EI	Electronics & Instrumentation Engineering
20.	ET	Telecommunication Engineering
21.	IEM	Industrial Engineering & Management
22.	IS	Information Science & Engineering
23.	ME	Mechanical Engineering
24.	CD	Computer Science & Engineering (Data Science)
25.	CY	Computer Science & Engineering (Cyber Security)
26.	CI	Computer Science & Engineering (Artificial Intelligence & Machine Learning)

Bachelor of Engineering in
INDUSTRIAL ENGINEERING AND MANAGEMENT

VII SEMESTER														
Sl. No.	Course Code	Course Title	Credit Allocation				BoS	Category	CIE Duration (H)	Max Marks CIE		SEE Duration (H)	Max Marks SEE	
			L	T	P	Total				Theory	Lab		Theory	Lab
1.	HS271TA	Indian Knowledge System	3	0	0	3	HS	Theory	1.5	100	--	3	100	--
2.	IM372IA	Product Design & Development	3	0	1	4	IM	Theory + Lab	1.5	100	50	3	100	50
3.	IM373TA	Total Quality Management	3	1	0	4	IM	Theory	1.5	100	--	3	100	--
4.	IM374TFX	Professional Core Elective-IV (Group-F)	3	0	0	3	IM	Theory	1.5	100	--	3	100	--
5.	XX375TGX	Institutional Electives – II (Group -G)	3	0	0	3	Resp. BoS	Theory	1.5	100	--	3	100	--
6.	IM376SI	Summer Internship	0	0	3	3	IM	Internship	---	--	100	3	--	100
						20								

VIII SEMESTER														
Sl. No.	Course Code	Course Title	Credit Allocation				BoS	Category	CIE Duration (H)	Max Marks CIE		SEE Duration (H)	Max Marks SEE	
			L	T	P	Total				Theory	Lab		Theory	Lab
1	IM481P	Major Project	0	0	12	12	IM	Project Work	----	---	100	3	---	100
						12								

Index

VII Semester			
Sl. No.	**Course Code**	**Course Title**	**Page No.**
1.	HS271TA	Indian Knowledge System	1 – 2
2.	IM372IA	Product Design & Development	3 – 5
3.	IM373TA	Total Quality Management	6 – 7
4.	IM374TFX	Professional Core Elective-IV (Group-F)	8 – 15
5.	XX375TGX	Institutional Electives – II (Group -G)	16 – 55
6.	IM376SI	Summer Internship	56 – 57

Professional Core Elective-IV (Group-F)

Sl. No.	Course Code	Course Title	Page No.
1	IM374TFA	Predictive Analytics	8 – 9
2	IM374TFB	Artificial Intelligence in Industrial and Systems Engineering	10 – 11
3	IM374TFC	Digital Supply Chain Management	12 – 13
4	IM374TFD	Sustainability Engineering and Circular Economy	14 – 15

Institutional Electives – II (Group -G)

Sl. No.	Course Code	BoS	Course Title	Page No.
1.	AI375TGA	AI	Data and Story Telling	16 – 17
2.	AS375TGB	AS	Aircraft Systems	18 – 19
3.	BT375TGC	BT	Health Care Technology for Engineers	20 – 21
4.	CH375TGD	CH	Green and Hydrogen Technology	22 – 23
5.	CM375TGE	CM	Chemistry of materials and molecular analysis	24 – 25
6.	CS375TGF	CS	Prompt Engineering	26 – 27
7.	CV375TGG	CV	Solid Waste Management and Statutory rules	28 – 29
8.	CV375TGH	CV	Freight Transportation Systems and Logistics	30 – 31
9.	EC375TGI	EC	IoT for smart systems	32 – 33
10.	EE375TGJ	EE	E-Mobility	34 – 35
11.	EI375TGK	EI	Disease and Diagnostics an Engineering Perspective	36 – 37
12.	ET375TGL	ET	Space Technology and Applications	38 – 39
13.	IM375TGM	IM	Project Management	40 – 41
14.	IM375TGN	IM	Global Supply chain management	42 – 43
15.	MA375TGO	MA	Statistical Methods for Engineers	44 – 45
16.	ME375TGP	ME	Industry 5.0	46 – 47
17.	PY375TGQ	PY	Principles of Astrophysics	48 – 49
18.	MA375TGR	MA	Mathematics of Music	50 – 51
19.	HS375TGS	HS	Cognitive Psychology	52 – 53
20.	HS375TGT	HS	Principles & Practices of Cyber Law	54 – 55

VII Semester			
Sl. No.	**Course Code**	**Course Title**	**Page No.**
1	IM481P	Major Project	58 – 59

Semester: VII						
INDIAN KNOWLEDGE SYSTEM **Common Course** **Theory**						
Course Code	**:**	**HS271TA**		**CIE**	**:**	**100 Marks**
Credits: L:T:P	**:**	**3:0:0**		**SEE**	**:**	**100 Marks**
Total Hours	**:**	**45 Hrs**		**SEE Duration**	**:**	**3 Hours**

Unit-I	**09 Hrs**
Introduction of Indian Knowledge Systems (IKS): Perception and prologue, Historicity of IKS, Indian Traditional Knowledge, Nature and scope, kinds of traditional knowledge, Transition from traditional to western knowledge. The IKS Corpus, Vedic Corpus, Indian Philosophical Systems.	
Unit – II	**09 Hrs**
Foundational Concepts of Mathematics and Science & Technology: Linguistics; Pānini's work on Sanskrit Grammar, Phoenetics, Ashtadhyayi's vyakarana. Number System and Units of Measurement: concept of zero, Piṅgala and the Binary system. Knowledge: Framework & Classifications: Prameya, Pramana, Samsaya.	
Unit –III	**09 Hrs**
Science, Engineering & Technology in IKS: Mathematics; Arithmetic, geometry, trigonometry and algebra, Chanda Sastra of Pingala. Astronomy: Elements of the Indian Calendar, Āryabhaṭīya and the Siddhāntic Tradition, Panchanga, Astronomical Instruments (Yantras) and Jyotish. Engineering & Technology: Metals & Metal Working; Wootz Steel, Iron and Steel in India, Metals and Metal working Technology, Lost-wax casting of idols and artefacts, Dyes and Painting Technology, The art of making perfumes.	
Unit –IV	**09 Hrs**
Yoga and Ayurveda: Tridoṣas, Triguṇa System, Body-Mind-Intellect-Consciousness Complex, Sixty-four art forms and occupational skills (64 Kalas). **Irrigation Systems and Water management Practices Town Planning & Architecture**: Indian Architecture in India, Vāstu Śāstra; Eight limbs of Vāstu, Town Planning; Temple Architecture	
Unit –V	**09 Hrs**
Governance, Public Administration & Management: Rāmāyaṇa on great attributes, Arthaśāstra – Governance & Administration, Janapada, Durga, Kosa, Danda, Mitra. IKS & United Nations Sustainable development goals. Safeguarding traditional Indian Knowledge.	

Course Outcomes: After completing the course, the students will be able to:-	
CO1	Explain the nature, scope, and historical development of Indian Knowledge Systems and differentiate traditional knowledge from modern scientific paradigms
CO2	Analyze the foundational concepts of Indian contributions to linguistics, mathematics, and science, including works of Pānini, Piṅgala and others
CO3	Evaluate the applications of traditional Indian technologies in areas such as astronomy, metallurgy, architecture, and medicine.
CO4	Assess the relevance of Indian Knowledge Systems in the context of sustainable development goals and propose methods to safeguard and integrate IKS in contemporary society

Reference Books	
1.	Introduction to Indian Knowledge System Concepts & Applications, B Mahadevan, Vinayaka Rajat Bhat, R N Nagendra Pavana, PHI Learning publishers, ISBN-13: 978-9391818203
2.	Traditional Knowledge System in India, Amit Jha, 2009, Atlantic Publishers and Distributors (P) Ltd., ISBN-13: 978-8126912230
3.	Knowledge Traditions and Practices of India, Kapil Kapoor, Avadesh Kumar Singh, Vol. 1, 2005, DK Print World (P) Ltd., ISBN 81-246-0334,
	Suggested Web Links:
1.	https://www.youtube.com/watch?v=LZP1StpYEPM
2.	http://nptel.ac.in/courses/121106003/
3.	http://www.iitkgp.ac.in/department/KS;jsessionid=C5042785F727F6EB46CBF432D7683B63 (Centre of Excellence for Indian Knowledge System, IIT Kharagpur)
4.	https://www.wipo.int/pressroom/en/briefs/tk_ip.html

5.	https://unctad.org/system/files/official-document/ditcted10_en.pdf
6.	http://nbaindia.org/uploaded/docs/traditionalknowledge_190707.pdf
7.	https://unfoundation.org/what-we-do/issues/sustainable-development-goals/?gclid=EAIaIQobChMInp-Jtb_p8gIVTeN3Ch27LAmPEAAYASAAEgIm1vD_BwE

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	**COMPONENTS**	**MARKS**
1.	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. **THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.**	**20**
2.	**TESTS:** Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom‟s Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks**. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) **Phase 2 will be done in the exhibition mode (Demo/Prototype/any outcome). ADDING UPTO 40 MARKS.**	**40**
	MAXIMUM MARKS FOR THE CIE (THEORY)	**100**

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	20
	PART B (Maximum of TWO Sub-divisions only)	
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
	TOTAL	**100**

Semester: VII						
PRODUCT DESIGN & DEVELOPMENT **Professional Core Course** **(Theory & Practice)**						
Course Code	**:**	**IM372IA**		**CIE**	**:**	**100 + 50 Marks**
Credits: L:T:P	**:**	**3:0:1**		**SEE**	**:**	**100 + 50 Marks**
Total Hours	**:**	**45L+ 30P**		**SEE Duration**	**:**	**3 + 3Hours**

Unit-I	**09 Hrs**
Introduction: Definition of product design, design by evolution, design by innovation, Essential factors of Product design, Characteristics of successful product development, Stage gate decision making, Design thinking approach to product design. **Development Processes and Organizations:** A generic development process, concept development: the front-end process, adapting the generic product development process.	
Unit – II	**09 Hrs**
Product Planning: The product planning process, identify opportunities. Product strategies, Analysis of a product. Definition of product design in the AI context. Design thinking with AI tools. Generative AI in front-end product development processes. **Identifying Customer Needs:** Steps in Identifying customer needs, Quality Function Deployment. AI sentiment analysis for customer needs. Machine learning for organizing and prioritizing data. **Product Specifications:** What are specifications, Basic design considerations and constraints, Various types of specification, Steps involved.	
Unit –III	**08 Hrs**
Concept Generation: The activity of concept generation, Steps involved, Generative AI-based concept ideation. **Concept Selection:** Overview of methodology, concept screening, concept scoring, caveats. Concept Testing, steps in concept testing, AI-assisted benchmarking and evaluations. **Product Architecture:** What is product architecture, implications of the architecture, establishing the architecture, variety and supply chain consideration.	
Unit –IV	**09 Hrs**
Industrial Design: Assessing the need for industrial design, the impact of industrial design, industrial design process, managing the industrial design process, assessing the quality of industrial design. AI-enhanced industrial design and creativity. **Ergonomics in Product Design:** Human-Machine Interaction principles. Applications of Anthropometry in design. Integration of Ergonomics in IoT and Smart Devices. AI simulations for ergonomic assessments. AI in cognitive ergonomics and workspace optimization.	
Unit –V	**09 Hrs**
Design for Manufacturing: Definition, Approach to design, Production Requirements, steps in DFM, Concurrent engineering, reasons for adopting concurrent engineering, AI-driven cost estimation for manufacturing processes. **Prototyping:** Prototyping basics, principles of prototyping, technologies, planning for prototypes. **Virtual Prototyping** for ergonomic evaluation.	
Laboratory	
Emphasis on Project based learning where group of students (max of 5) will be expected to develop designs of product and develop prototypes by using tentative exercises mentioned below. • Scouting for product idea- Visit to any related facility in the vicinity (Cross over learning) • Value chain analysis and VRIO Framework • Project Charter using Notion.IO and Work Breakdown Structure • Brain storming and Mind mapping of futuristic products • Crazy 8's method, SCAMPER, Brain writing • Concept Generation using Morph matrix and Concept selection using Pugh matrix • Mom test and Monkey test • S/W tools used by Product managers (Clickup, Figma, Miro, Product board)- Any two to be explored. • Demonstration of Engine assembly- Mercedes lab/ MG lab/ Wirin lab/ Toyota Lab	

Course Outcomes	
CO1	Explain the structured approaches to Product design and development projects.
CO2	Understand the challenges facing product designers and appreciate the need for adapting a development mind set
CO3	Develop the capability to work in teams and apply the structured product design and development methodologies for solving problems.
CO4	Analyze the need for integrated product design and process development frameworks.

Reference Books	
1.	Product Design and Development, Karl. T. Ulrich, Steven D Eppinger, edition 2009, 2008, Tata McGrawHill, ISBN – 0-07058513-X.
2.	Product Design and Manufacturing, A C Chitale and R C Gupta, 4th Edition, 2007, PHI, ISBN: 9788120333178.
3.	Product Design, Kevin Otto and Kristen Wood, 1st edition, 2001, Pearson Education-, ISBN-10: 0130212717.
4.	Introduction to Ergonomics, R S Bridger, 4th Edition, 2017, CRC Press, ISBN: 9781498795944.
5	Human Factors in Engineering and Design; Mark S. Sanders and Ernest J McCormick; 7th Edition, McGraw-Hill Education, 2013, ISBN-10: 1259064727 /13: 978-1259064722.

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	**COMPONENTS**	**MARKS**
1.	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. **THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.**	**20**
2.	**TESTS:** Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). THREE tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 150 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) **ADDING UPTO 40 MARKS**.	**40**
4.	LAB: Conduction of laboratory exercises, lab report, observation, and analysis (20 Marks), lab test (10 Marks) and Innovative Experiment/ Concept Design and Implementation (20Marks) adding up to 50 Marks. **THE FINAL MARKS WILL BE 50 MARKS**	**50**
	MAXIMUM MARKS FOR THE CIE (THEORY AND PRACTICE)	**150**

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	20
	PART B (Maximum of TWO Sub-divisions only)	
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
	TOTAL	**100**

RUBRIC FOR SEMESTER END EXAMINATION (LAB)		
Q. NO.	**CONTENTS**	**MARKS**
	PART A	
1	Write Up	10
2	Conduction of the Experiments	20
3	Viva	20
	TOTAL	**50**

Semester: VII						
TOTAL QUALITY MANAGEMENT **Professional Core Course** **(Theory)**						
Course Code	**:**	**IM373TA**		**CIE**	**:**	**100 Marks**
Credits: L:T:P	**:**	**3:1:0**		**SEE**	**:**	**100 Marks**
Total Hours	**:**	**45L+ 15T**		**SEE Duration**	**:**	**3 Hours**

UNIT-I	**09 Hrs**
Quality Pioneers: Deming's approach, Juran's quality trilogy, Crosby and quality treatment, Imai's Kaizen, Ishikawa's company-wide quality control, and Feigenbaum's theory of TQC. **Evolution of Quality Concepts and Methods:** Quality concepts, Development of four fitness's, evolution of methodology, evolution of company integration. ISO 56000, **Business Excellence Models:** MBQNA, EFQM, Deming's awards, Rajiv Gandi Golden Peacock award.	
UNIT-II	**09 Hrs**
Four Revolutions in Management thinking, Focus on customers: Change in work concept, market-in, and customers. Continuous Improvement: Improvement as problem solving process: Management by process, WV model of continuous improvement. **Reactive Improvement:** Identifying the problem, standard steps, seven steps case study, General guidelines for managers diagnosing a QI story, 8D, 12D approaches. **Proactive Improvement:** Introduction to proactive improvement, standard steps for proactive improvement, semantics, Seven Management and Planning Tools.	
UNIT-III	**09 Hrs**
Total Participation: Teamwork skill, Dual function of work, teams and teamwork, principles for activating teamwork, creativity in team processes, Initiation strategies, **Hoshin Management:** Definition, Concepts, Phases in Hoshin Management – overview. Societal **Networking:** Networking and societal diffusion, infrastructure for networking. TQM as learning system, a TQM model for skill development.	
UNIT-IV	**09 hrs**
A Very Brief History of the Quality Movement, Defining Quality for Small Business, Six Sigma and Your Employees Six Sigma and Company Culture The Role of the Small Business Owner/Manager, Progress Assessment, Key Six Sigma Players, Executive Leader(s), Champions ,Black Belts Master Black Belt, Green Belts, Key Criteria for Project Selection, The Low-Hanging vs. Rotted Fruit, Scoping a Project, Project Ideas, Critical-to-Quality, The Project Problem Statement, Six Steps to Six Sigma.	
UNIT-V	**09 Hrs**
Design for Six Sigma: Overview of DFSS, DMADV Method **Your Six Sigma Project:** The Define Phase, The Measure Phase, The Analyze Phase, The Improve Phase, The Control Phase	

Course Outcomes	
CO1	Explain the TQM & Six Sigma principles and concepts for organizations
CO2	Compare TQM and Six Sigma methodologies.
CO3	Evaluate and select the appropriate framework for continuous improvement
CO4	Design & implement TQM & Six Sigma projects in organizational situations

Reference Books	
1.	Shoji Shiba, Alan Graham and David Walden, A New American TQM – Four Practical Revolutions in Management, Productivity Press, Portland (USA), 2nd Edition, 1993, ISBN: 9781563270321
2.	Six sigma for small business by Greg Brue .p. cm. ISBN 1-932531-55-6 (alk. paper) HD62.7.B79 2005,658.4'013--dc22
3.	N Logothetis , Managing for total quality: from Deming to Taguchi and SPC, Prentice Hall of India,1993, ISBN: 978-0133535127
4.	Dale H. Besterfield, Carol Besterfield-Michna, Glen Besterfield, Mary Besterfield – Sacre, Total Quality Management, Pearson Education, 2002, 3rd Edition, ISBN-81-297-0260-6.
5.	Greg Brue and Rod Howes, Six Sigma, TATA McGraw-Hill Edition 2006, ISBN: 0-07-063468-8

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	**COMPONENTS**	**MARKS**
1.	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. **THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.**	**20**
2.	**TESTS:** Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). THREE tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 150 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20**) ADDING UPTO 40 MARKS**.	**40**
	MAXIMUM MARKS FOR THE CIE (THEORY AND PRACTICE)	**100**

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	20
	PART B (Maximum of TWO Sub-divisions only)	
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
	TOTAL	**100**

Semester: VII						
PREDICTIVE ANALYTICS **Professional Core Elective – IV (Group F)** **(Theory)**						
Course Code	**:**	**IM374TFA**		**CIE**	**:**	**100 Marks**
Credits: L:T:P	**:**	**3:0:0**		**SEE**	**:**	**100 Marks**
Total Hours	**:**	**45L**		**SEE Duration**	**:**	**3 Hours**
Unit-I						**09 Hrs**
Introduction to Data Mining: Overview of data mining concepts and process, Data preprocessing techniques (cleaning, transformation, reduction), Exploratory data analysis, Introduction to data mining software/tools (e.g., R, Python libraries)						
Unit – II						**09 Hrs**
Classification and Prediction: Classification algorithms (decision trees, logistic regression, naïve Bayes), Evaluation of classification models (accuracy, precision, recall), Predictive modeling and regression analysis, Application of classification and prediction algorithms in industrial engineering and management scenarios						
Unit –III						**09 Hrs**
Clustering and Association Analysis: Clustering algorithms (k-means, hierarchical clustering), Association analysis (Apriori algorithm), Interpretation and evaluation of clustering and association patterns, Case studies on supply chain optimization and customer segmentation						
Unit –IV						**09 Hrs**
Advanced Topics in Data Mining: Ensemble methods (random forests, gradient boosting), Feature selection and dimensionality reduction techniques, Text mining and sentiment analysis, Anomaly detection, Application of advanced data mining techniques in industrial engineering and management domains						
Unit –V						**09 Hrs**
Practical Applications and Case Studies: Integration of data mining techniques into industrial engineering and management practices, Real-world case studies in manufacturing, logistics, finance, and healthcare sectors, Discussion on ethical considerations, privacy concerns, and responsible use of data mining in industry, Data Visualization, Dashboard.						

Course Outcomes	
CO1	Demonstrate a comprehensive understanding of fundamental concepts in data mining and predictive analytics.
CO2	Apply a variety of data mining techniques such as classification, regression, clustering, and association rule mining to real-world datasets.
CO3	Evaluating the performance of predictive models using techniques such as cross-validation, Receiver-operating characteristic (ROC) analysis, and confusion matrices.
CO4	Implement end-to-end predictive analytics solutions, including data preprocessing, model selection and training, validation, and deployment

Reference Books	
1.	Data Mining and Predictive Analysis, Daniel T Larose, 2nd Edition, 2015, Wiley, ISBN 978-1-118-11619-7
2.	Data Mining for Business Analytics, Shmueli, Bruce, Yahav, Patel, Lichtendahl Jr. adapted by O P Wali, 2021, Wiley, ISBN: 978-93-90421-70-1
3.	Applied Predictive Analytics: Principles and Techniques for The Professional Data Analyst, Dean Abbott, 2014, John Wiley and Sons, ISBN: 978-1-118-72796-6
4.	Fundamentals of Machine Learning for Predictive Data Analytics, John D. Kelleher, Brian Mac Namee and Aoife D'Arcy, 2015, MIT Press, ISBN: 97802620 29445

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	COMPONENTS	MARKS
1.	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. **THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.**	**20**
2.	**TESTS:** Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). THREE tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 150 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) **ADDING UPTO 40 MARKS**.	**40**
	MAXIMUM MARKS FOR THE CIE (THEORY AND PRACTICE)	**100**

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	20
	PART B (Maximum of TWO Sub-divisions only)	
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
	TOTAL	**100**

Semester: VII						
ARTIFICIAL INTELLIGENCE IN INDUSTRIAL AND SYSTEMS ENGINEERING **Professional Core Elective – IV (Group F)** **(Theory)**						
Course Code	**:**	**IM374TFB**		**CIE**	**:**	**100 Marks**
Credits: L:T:P	**:**	**3:0:0**		**SEE**	**:**	**100 Marks**
Total Hours	**:**	**45L**		**SEE Duration**	**:**	**3 Hours**

UNIT-I	**09 Hrs**
Introduction to AI and Machine Learning for Industrial Engineering (IE): Introduction to Artificial Intelligence (AI): Foundations, History, Evolution, and The state of the art. Fundamentals of Machine Learning (ML): A General Model of Learning Agents, Inductive Learning, Learning Decision Trees, Using Information Theory. Traditional IE and AI enabled IE, The Role of AI in Industrial and Systems Engineering: Enhancing Efficiency, Productivity, and Decision-Making.	
UNIT – II	**09Hrs**
AI for Operations and Supply Chain Traditional Optimization Techniques vs. AI-Powered Optimization, AI/ML for Demand Forecasting and Inventory Management, AI-based Approaches for Production Scheduling and Resource Allocation, Applications in Supply Chain Optimization and Logistics. Case Studies: AI in Manufacturing Scheduling, Warehouse Optimization.	
UNIT –III	**09Hrs**
AI-driven Quality Inspection and Defect Detection Predictive Maintenance using Machine Learning: Concepts and Techniques, Anomaly Detection for Equipment Health Monitoring, Root Cause Analysis using AI techniques, Case Studies: AI in Automotive Quality Control, Condition Monitoring in Aerospace.	
Unit –IV	**09Hrs**
AI for Human Factors and Ergonomics AI-powered analysis of human motion and task performance, Ergonomic Risk Assessment using Computer Vision and AI, Intelligent Assistance Systems and Human-Robot Collaboration, AI for personalized training and skill development in industrial settings. Case Studies of AI for Human Factors and Ergonomics	
UNIT –V	**09 Hrs**
Unit 5: Emerging Trends and Implementation Challenges AI for Sustainability and Circular Economy in Industrial Operations, Ethical & Safety Considerations and Responsible AI in Industrial & Systems Engineering, Challenges and Barriers to AI Adoption in Industries, Future Trends and Research Directions in AI for Industrial Engineering.	

Course Outcomes	
CO1	To explain fundamental AI techniques applicable in industrial engineering.
CO2	To analyse how AI can enhance productivity, quality, and decision-making in industrial engineering
CO3	To Identify and analyze potential applications of AI in various areas of industrial engineering.
CO4	To evaluate the benefits and challenges of implementing AI solutions in industrial engineering applications

Reference Books	
1.	Stuart Russell and Peter Norvig, Artificial Intelligence: A Modern Approach, 4th Edition, Prentice Hall, 2022, ISBN-10 : 9356063575, ISBN-13 : 978-9356063570
2.	Tom M. Mitchell, Machine Learning,, McGraw Hill, 1st Edition, 2017, SBN-10 : 1259096955, ISBN-13 : 978-1259096952
3.	Aurélien Géron, Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems, 3rd Edition, 2022, ISBN-10 : 9355421982, ISBN-13 : 978-9355421982
4.	Zhen "Leo" Liu, Artificial Intelligence for Engineers – Basics and Implementation, Springer Nature Link, 1st Edition, 2025, ISBN 978-3-031-75952-9

Research Papers and Articles in journals such as:

- International Journal of Production Research
- Computers & Industrial Engineering
- Journal of Manufacturing Systems
- IEEE Transactions on Automation Science and Engineering
- Artificial Intelligence in Engineering

Conference Proceedings from conferences like:

- IISE Annual Conference
- International Conference on Industrial Engineering and Operations Management (IEOM)
- IEEE International Conference on Robotics and Automation (ICRA)
- Neural Information Processing Systems (NeurIPS)
- International Conference on Machine Learning (ICML)

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	**COMPONENTS**	**MARKS**
1.	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. **THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.**	**20**
2.	**TESTS:** Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). THREE tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 150 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20**) ADDING UPTO 40 MARKS**.	**40**
	MAXIMUM MARKS FOR THE CIE (THEORY AND PRACTICE)	**100**

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	20
	PART B (Maximum of TWO Sub-divisions only)	
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
	TOTAL	**100**

Semester: VII						
DIGITAL SUPPLY CHAIN MANAGEMENT **Professional Core Elective – IV (Group F)** **(Theory)**						
Course Code	**:**	**IM374TFC**		**CIE**	**:**	**100 Marks**
Credits: L:T:P	**:**	**3:0:0**		**SEE**	**:**	**100 Marks**
Total Hours	**:**	**45L**		**SEE Duration**	**:**	**3 Hours**

Unit-I	**09 Hrs**
The Digital Supply Chain—emergence, concepts, definitions, and technologies*:* A transformative decade, Emergence of the Digital Supply Chain, Building Blocks for the Digital Supply Chain, Defining the Digital Supply Chain, Opportunities, Challenges	
Unit – II	**09 Hrs**
Smart warehouses—a sociotechnical perspective - The digital supply chain transforms the requirements for warehousing, Warehouse management, Smart warehouses: enabling technologies, Order-picking in the smart warehouse, Real-time location tracking using IoT, IoT-enabled AS/RS systems **The impact of digitalization on contemporary and future logistics**: Introduction, Digitalization in logistics and supply chain management, Cloud-based systems, Drone Technologies, Emerging technologies	
Unit –III	**09 Hrs**
Blockchain technologies in the digital supply chain: Introduction, Functionality of blockchain, Blockchain in the academic supply chain literature, Industrial applications of blockchain. **Digital architectures: frameworks for supply chain data and information governance:** Introduction, Data as a resource—the need for data quality, Data and information architectures, Data sharing agreements, Data attributes, sharing, and access control, Actors, roles, and relationships in data sharing	
Unit –IV	**09 Hrs**
Supply chain traceability systems—robust approaches for the digital age: Introduction, Visibility, transparency, and traceability, Motivations for traceability and transparency, Information requirements for traceability systems, Enabling technologies, Challenges, **Digital purchasing and procurement systems: evolution and current state** - Introduction—the rise of digital procurement systems, the development of digital procurement systems, Looking ahead: the future of digital procurement systems	
Unit –V	**09 Hrs**
Measuring and managing digital supply chain performance: Introduction, A framework for performance management in digital supply chains, Case studies, Impact of emerging technologies on performance measurement and management **The art of cyber security in the age of the digital supply chain: detecting and defending against vulnerabilities in your supply chain,** Introduction, Governments, consultancies, and industry approaches	

Course Outcomes	
CO1	Understand the digital supply chain concepts, technologies, and implications, enabling them to analyze, strategize, and innovate within the evolving landscape of supply chain management.
CO2	Analyze the significance of smart warehouses and assess the effects of digitalization on logistics operation
CO3	Comprehend block chain technologies and digital architectures propose innovative solutions for enhanced efficiency and security in digital environments.
CO4	Use digital tools for traceability, procurement, performance evaluation, and vulnerability management in the digital supply chain.

Reference Books	
1.	The Digital Supply Chain, Bart L. MacCarthy, Dmitry Ivanov, 1st Edition - June 9, 2022 Elsevier, ISBN: 9780323916141
2.	Digital Supply Chains: A Practitioner's Guide to Successful Digitalization, Mrozek, Thomas, et al., 2020, Germany, Campus Verlag, ISBN: 9783593443973

3.	Digital Supply Networks: Transform Your Supply Chain and Gain Competitive Advantage with Disruptive Technology and Reimagined Processes, Sinha, A., Bernardes, E., Calderon, R., Wuest, T. 2020, McGraw Hill LLC, ISBN:9781260458206

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	**COMPONENTS**	**MARKS**
1.	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. **THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.**	**20**
2.	**TESTS:** Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). THREE tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 150 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) **ADDING UPTO 40 MARKS**.	**40**
	MAXIMUM MARKS FOR THE CIE (THEORY AND PRACTICE)	**100**

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	20
	PART B (Maximum of TWO Sub-divisions only)	
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
	TOTAL	**100**

Semester: VII						
SUSTAINABILITY ENGINEERING AND CIRCULAR ECONOMY **Professional Core Elective – IV (Group F)** **(Theory)**						
Course Code	**:**	**IM374TFD**		**CIE**	**:**	**100 Marks**
Credits: L:T:P	**:**	**3:0:0**		**SEE**	**:**	**100 Marks**
Total Hours	**:**	**45L**		**SEE Duration**	**:**	**3 Hours**

UNIT-I	**09 Hrs**
Basics of Sustainable Engineering: Introduction, Why and What is Sustainable Development, THE SDGs, Paris Agreement and Role of Engineering, Sustainable Development and the Engineering Profession, Key attributes of the Graduate Engineering **Concepts and Principles of Sustainable Engineering:** Goals of sustainability, System Thinking, Life Cycle Thinking and Circular Economy, Green Economy and Low Carbon Economy, Eco Efficiency, Triple bottom Line, Guiding principles of sustainable engineering, Frameworks for sustainable Engineering.	
UNIT – II	**09Hrs**
Fundamentals of Life Cycle Assessment: Why and What is LCA, LCA Goal and Scope, Life cycle inventory, Life Cycle Impact Assessment, Interpretation and presentation of Results, Iterative Nature of LCA, Methodological Choices, LCI Databases and LCA Software's, Strength and Limitations of LCA. Social Impact & Life Cycle Costing, Life Cycle Cost Analysis tools **Life Cycle Assessment for Decision Making:** How LCA helps in Decision Making, Future Prospects and Sustainable Assessment.	
UNIT –III	**09Hrs**
Engineering Sustainability, Needs for Metrology and Standards: Introduction, Providers of Standards and Methods of Development, Standards, A Conceptual Standard Framework, Corporate Sustainability **Sustainability and Future Trends:** Introduction, Sustainable Operations and Supply Chain Management, The Impact of Climate Change, Demographic Evolution, Urbanization, Emerging New Consumers, Smart Technologies and the Digital Factory, Mega-Trends,	
Unit –IV	**09Hrs**
Introduction to Circular Economy: Towards Circular Economy, What's the Need. Principles of Circular Economy, Understanding the Concepts, Framework, Design, UN SDGs & Circular Economy **Circular Economy and Engineering:** Integrating Sustainable Engineering, Circular Economy in Engineering Discipline, Circular Eco -Design (3Rs: Re -think, Re -Design & Re -Purpose), Circular Eco Design using the 7R model	
UNIT –V	**09 Hrs**
Circular Economy and Environment: Impact on Environment Policies, Aids in Resource Management, Impact on Economic and Social Indicators, Advantages of Circular economy than Linear Economy, Economy and Sustainability **Sectoral and Business Case studies**: Sectoral and Business Case studies of Circular Economy a. Case Study -1 (Environment) b. Case Study -2 (Developed Country) c. Case Study -3 (Developing Country) d. Case Study -4 (Industry)	

Course Outcomes	
CO1	Interpret sustainable engineering principles and the role of engineers in achieving global sustainability goals.
CO2	Apply Life Cycle Assessment to evaluate environmental and economic impacts of engineering systems.
CO3	Analyze sustainability trends to assess their influence on operations and supply chains.
CO4	Integrate circular economy principles into engineering design and decision-making.

Reference Books	
1.	Introduction to Sustainability for Engineers, by Toolseeram Ramjeawon. CRC Press; 1st edition (Feb. 28 2020). ISBN-10: 036725445X;
2.	Practicing Circular Economy, by Prasad Modak, CRC Press, ISBN 9780367619572.
3.	Sustainable operations and supply chain management, Belvedere, Valeria, and Alberto Grando. John Wiley & Sons, 2017. ISBN: 9781119285366, 1119285364
4.	Measuring Progress Towards Sustainability - A Treatise for Engineers, Subhas K. Sikdar, Debalina Sengupta, Rajib Mukherjee, ebook, Springer International Publishing, ISBN : 9783319427195, 3319427199

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	COMPONENTS	MARKS
1.	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. **THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.**	**20**
2.	**TESTS:** Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). THREE tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 150 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) **ADDING UPTO 40 MARKS**.	**40**
	MAXIMUM MARKS FOR THE CIE (THEORY AND PRACTICE)	**100**

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	20
	PART B (Maximum of TWO Sub-divisions only)	
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
	TOTAL	**100**

Semester: VII						
DATA AND STORYTELLING **Institutional Elective – II (Group G)** **(Theory)**						
Course Code	**:**	**AI375TGA**		**CIE**	**:**	**100 Marks**
Credits: L:T:P	**:**	**3:0:0**		**SEE**	**:**	**100 Marks**
Total Hours	**:**	**45L**		**SEE Duration**	**:**	**3Hours**

Unit-I	**09 Hrs**
Understanding Data: Types, Sources, and Preparation: What is data and why it matters, Types of data – qualitative, quantitative, structured, unstructured, 3Vs of big data – volume, velocity, variety, Common data sources – government portals, sensors, APIs, crowd sourced platforms, Data representations – tabular, relational, JSON, XML formats, Introduction to spreadsheets and data frames, Basics of data preprocessing – missing value handling, normalization, type casting, cleaning workflows using Python (pandas) or spreadsheets	
Unit – II	**09 Hrs**
Data Collection for Exploratory Research: Methods of primary data collection – surveys, interviews, FGDs, observations, Designing structured questionnaires and interview guides, Ensuring question clarity, neutrality, and goal alignment, Sampling strategies – random, stratified, purposive sampling, Field planning and respondent communication, Ethical considerations – consent, privacy, data sensitivity, Transcription and textual data cleaning.	
Unit –III	**09 Hrs**
Sensor Data Collection and logging: Introduction to IoT and embedded data logging, Sensor interfacing – temperature, humidity, motion, light sensors, Analog-to-digital conversion and signal conditioning, Sampling rates and real-time clock synchronization, Wireless transmission – BLE, Wi-Fi, LoRa, Local and remote data logging – SD cards, cloud platforms, Handling data quality issues – noise, missing values, power constraints, security and privacy (Man in the middle attacks). Real-time plotting, Visualizing trends, anomalies, and multi-sensor overlays, Debugging and optimizing embedded data pipelines, Mapping Urban Tree Inequality: A Data-Driven Look at Green Access, Building a Local Data Chain to Map Green Equity in Our Cities, Project Pakshi	
Unit –IV	**09 Hrs**
Storytelling through Data: Communicating patterns and trends with clarity, Building simple, reproducible analysis notebooks, Introduction to data storytelling and presentation, Structuring analysis reports – insights, visuals, limitations, Creating visual narratives – dashboards, summaries, infographics, Ethical data interpretation – avoiding bias and misrepresentation	
Unit –V	**09 Hrs**
Statistics Done Wrong: Statistical Significance, Statistical Power, and the Problem of Underpowered Studies, Pseudoreplication, Model Abuse, Researcher Freedom, Hiding the data. Final project – design, collect, visualize, and present data using field instruments, embedded systems, or secondary sources	

Course Outcomes: After completing the course, the students will be able to:-	
CO1	Understand different types, sources, and representations of data, and perform basic preprocessing using modern tools.
CO2	Design and implement primary data collection strategies using surveys, interviews, and sampling techniques with ethical considerations.
CO3	Interface and configure sensors for real-time data logging, transmission, and visualization while addressing data quality and privacy issues.
CO4	Create structured analysis reports and visual narratives to communicate insights clearly and ethically using dashboards and notebooks.
CO5	Identify and critique common statistical errors and misinterpretations in data analysis to promote valid and ethical research practices.

Reference Books	
1.	The Craft of Research, Wayne C. Booth, Gregory G. Colomb, and Joseph M. Williams, University of Chicago Press, Fifth Edition, 2024, ISBN-13: 978-0226826677
2.	Sensor Data Analysis and Management: The Role of Deep Learning, Editors: A. Suresh, R. Udendhran, M. S. Irfan Ahmed, Wiley-IEEE Press, 2021, ISBN-13: 978-1119682424

3.	Storytelling with Data: A Data Visualization Guide for Business Professionals, Cole Nussbaumer Knaflic, Wiley Publications, 2015, ISBN-13: 978-1119002253
4.	Statistics Done Wrong: The Woefully Complete Guide, Alex Reinhart, No Starch Press, 2015, ISBN-13: 978-1593276201

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	**COMPONENTS**	**MARKS**
1.	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. **THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.**	**20**
2.	**TESTS:** Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). **TWO tests will be conducted**. Each test will be evaluated for **50Marks**, adding upto 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) **ADDING UPTO 40 MARKS**.	**40**
	MAXIMUM MARKS FOR THE CIE THEORY	**100**

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	20
	PART B (Maximum of TWO Sub-divisions only)	
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
	TOTAL	**100**

Semester: VII						
AIRCRAFT SYSTEMS **INSTITUTIONAL ELECTIVE – II (GROUP G)** **(Theory)**						
Course Code	**:**	**AS375TGB**		**CIE**	**:**	**100 Marks**
Credits: L:T:P	**:**	**3:0:0**		**SEE**	**:**	**100 Marks**
Total Hours	**:**	**45L**		**SEE Duration**	**:**	**3.00 Hours**

Unit-I	**09 Hrs**
Flight Control Systems: History of Airplanes, Anatomy of an aircraft, Anatomy of an aircraft, Primary and secondary flight control surfaces, Flight control linkage system, Conventional Systems, Power assisted and fully powered flight controls, Electronic flight control system-fly by wire system.	
Unit – II	**09 Hrs**
Aircraft Hydraulic & Pneumatic Systems: Components of a typical Hydraulic system, Working or hydraulic system, Power packs, Hydraulic actuators, Pressure regulating devices, Hydraulic pumps, Pneumatic system and components, Use of bleed air, Landing gear and braking,	
Unit –III	**09 Hrs**
Aircraft Fuel Systems: APU, RAT, Characteristics of aircraft fuel system, Fuel system and its components, different types of fuel tanks, types of fuel lines, Fuel pumps-classification, Fuel control unit, Gravity feed and pressure feed fuel systems,	
Unit –IV	**09 Hrs**
Environmental Control Systems: Air-conditioning system, vapour cycle system, deicing and anti-icing system, Fire detection- warning and suppression. Crew escape aids. **Engine Systems:** Engine starting sequence, Starting and Ignition systems, Engine oils and a typical lubricating system.	
Unit –V	**09 Hrs**
Aircraft Instruments: Instruments displays, Instrumentation grouping, Navigation instruments, Radio instruments, Hydraulic and Engine instruments. **Air Data Instruments:** Basic air data system and probes, Mach meter, Air speed indicator, Vertical speed indicator, Barometric pressure sensing, Altimeter, Air data alerting system- angle of attack sensing, stall warning, Mach warning, altitude alerting system.	

Course Outcomes: After completing the course, the students will be able to:	
CO1	Demonstrate a comprehensive understanding of the fundamental components and operational principles of major aircraft systems.
CO2	Analyze and explain the functions of various subsystems of aircraft in emphasizing their roles in safe and efficient flight operations.
CO3	Apply knowledge of aircraft systems to identify typical issues and malfunctions related to flight controls and other subsystems
CO4	Evaluate the the regulatory considerations in aircraft system technologies for different flight envelopes

Reference Books	
1	Introduction to Flight, John D. Anderson, 7th Edition, 2011, McGraw-Hill Education, ISBN 9780071086059.
2	Fundamentals Of Flight Vol 4: Aircraft Systems, Lalit Gupta, Op Sharma, Himalayan Books, ISBN-13: 9788170020974
3	Flight stability and automatic control, Nelson R.C, 2nd Edition, 1998, McGraw-Hill International Editions, ISBN 9780071158381.
4	Aircraft Systems: Mechanical, Electrical and Avionics Subsystems Integration, Ian Moir, Allan Seabridge, 3rd Edition, 2008, John Wiley & Sons,. ISBN 978111965006.

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY & PRACTISE)		
#	**COMPONENTS**	**MARKS**
1.	**QUIZZES:** Quizzes will be conducted in online/offline mode. **TWO QUIZZES** will be conducted & Each Quiz will be evaluated for 10 Marks adding up to 20 Marks. **THE SUM OF TWO QUIZZES WILL BE CONSIDERED AS FINAL QUIZ MARKS.**	**20**
2.	**TESTS:** Students will be evaluated in test consisting of descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). **TWO TESTS** will be conducted. Each test will be evaluated for 50 Marks, adding up to 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. **Phase I (20) & Phase II (20) ADDING UPTO 40 MARKS**.	**40**
MAXIMUM MARKS FOR THE CIE (THEORY)		**100**

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO	**CONTENTS**	**MARKS**
PART A		
1	Objective type questions covering entire syllabus	**20**
	PART B (Maximum of THREE Sub-divisions only)	
2	Unit 1: (Compulsory)	**16**
3 & 4	Unit 2: Question 3 or 4	**16**
5 & 6	Unit 3: Question 5 or 6	**16**
7 & 8	Unit 4: Question 7 or 8	**16**
9 & 10	Unit 5: Question 9 or 10	**16**
	TOTAL	**100**

Semester: VII						
HEALTHCARE TECHNOLOGY FOR ENGINEERS **Institutional Elective – II (Group G)** **(Theory)**						
Course Code	**:**	**BT375TGC**		**CIE**	**:**	**100 Marks**
Credits: L:T:P	**:**	**3:0:0**		**SEE**	**:**	**100 Marks**
Total Hours	**:**	**45Hrs**		**SEE Duration**	**:**	**3Hours**

Unit-I	**09 Hrs**
Introduction to Healthcare Technology: National Digital Health Mission & Healthcare, IT Infrastructure, Telemedicine and remote healthcare applications, Basic Terminology and Role of Engineers in Healthcare Technology; Definition and importance of healthcare technology, Roles and responsibilities of engineers in the healthcare sector, Interdisciplinary collaboration between healthcare professionals and engineers, Key technologies used in modern healthcare systems.	
Unit – II	**09 Hrs**
Healthcare Data Management and Digital Epidemiology: Introduction to healthcare data types (structured, unstructured, real-time data), EHR in healthcare and its applications, Data security and privacy in healthcare, Role of digital epidemiology in tracking and predicting diseases, Use of AI and machine learning in healthcare data analysis.	
Unit –III	**09 Hrs**
Technology-Led Healthcare: Clinical Decision Support Systems (CDSS) and AI in Diagnosis Machine Learning and Deep Learning in Healthcare, Patient Monitoring Systems (PMS) and IoT Applications, Wearable Devices for Health Tracking. **Technology-Led Healthcare:** Quality Management Systems in Healthcare, Infection Prevention and Waste Management Technologies, Medical Device Development and Quality Assurance, Smart Hospitals: Case Study of a Technology-Enabled Hospital.	
Unit –IV	**09 Hrs**
Regulatory, Policy and Cybersecurity Aspects: Legal, Ethical and Intellectual Property Rights in Healthcare Technology, Cybersecurity in Healthcare; Data Protection and Privacy, Health Technology Assessment and Procurement Strategies.	
Unit –V	**09 Hrs**
Innovations in Healthcare Technology: Simulation and Organ Modelling for Medical Applications,3D Printing and Tissue Engineering in Healthcare, Augmented Reality (AR) and Virtual Reality (VR) in Medical Applications, Robotics and Biomechanics in Healthcare, Emerging Digital Health Technologies and Future Trends.	

Course Outcomes: After completing the course, the students will be able to:-	
CO1	Design, and implement healthcare technologies effectively
CO2	Analyse the perspective of sensing and imaging in technology for the better health
CO3	Ensuring compliance with regulatory and cybersecurity frameworks of healthcare data.
CO4	Evaluate various tools, techniques and advances for better formulation and productivity

Reference Books	
1.	"Biomedical Engineering: Bridging Medicine and Technology." Saltzman, W. Mark., Cambridge University Press, 2015. https://doi.org/10.1017/CBO9781139583831.
2.	"Medical Instrumentation: Application and Design.",Webster, John G., 4th ed., John Wiley & Sons, 2010.
3.	"Artificial Intelligence in Healthcare". Mahajan, Parag., 2nd (General) ed., Updated 2022.
4.	"Healthcare Information Technology Exam Guide for CHTS and CAHIMS Certifications". McCormick, Karen A., Barbara Gugerty, and James E. Mattison., McGraw Hill Professional, 2017.
5.	"Introduction to Biomedical Engineering". Enderle, John D., and Joseph D. Bronzino, editors.,3rd ed., Academic Press, 2012. ISBN: 978-0-12-374979-6.
6.	"Healthcare Information Privacy and Security: Regulatory Compliance and Data Security in the Age of Electronic Health Records". Robichau, Bernard Peter, Apress, 2014.

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	COMPONENTS	MARKS
1.	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. **THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.**	**20**
2.	**TESTS:** Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20**) ADDING UPTO 40 MARKS**.	**40**
MAXIMUM MARKS FOR THE CIE THEORY		**100**

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	20
	PART B (Maximum of TWO Sub-divisions only; wherein one sub division will be a caselet in the related topics)	
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16

Semester: VII						
GREEN AND HYDROGEN TECHNOLOGY **Institutional Elective – II (Group G)** **(Theory)**						
Course Code	**:**	**CH375TGD**		**CIE**	**:**	**100 Marks**
Credits: L:T:P	**:**	**3:0:0**		**SEE**	**:**	**100 Marks**
Total Hours	**:**	**45L**		**SEE Duration**	**:**	**3 Hours**

Unit-I	**09 Hrs**
Hydrogen: Peculiarity and Types Salient features of hydrogen, properties of hydrogen, terminology and types of hydrogen, advantages, disadvantages, comparison with other fuels, and global status of supply and demand.	
Unit – II	**09 Hrs**
Hydrogen Generation Generation of different types of hydrogen, conventional methods (generation from non-renewable sources), nonconventional methods (generation from renewable sources).	
Unit –III	**09 Hrs**
Hydrogen Storage Storage as compressed gas, storage as cryogenic liquid, storage as metal hydrides, storage through liquid organic hydrogen carriers, and other storage methods.	
Unit –IV	**09 Hrs**
Hydrogen Handling and Safety Classification of hydrogen hazards, compressed and liquid hydrogen related hazards, regulation, codes and standards related to hydrogen handling and transport.	
Unit –V	**09 Hrs**
Hydrogen Applications Applications of hydrogen in various sectors such as refineries, petrochemicals, fertilizer industries, transport and automotive sectors, steel industries, and AI in hydrogen technology.	

Course Outcomes	
CO1	Understand the importance of hydrogen and its use as an energy carrier
CO2	Explain the production, storage and handling of hydrogen
CO3	Analyze the need for hydrogen as an alternate fuel and the associated challenges
CO4	Appraise the importance of safety, regulations and codes

Reference Books	
1.	Hydrogen Safety: Production, Transport, Storage, Use, and the Environment, Fotis Rigas, CRC Press, Taylor & Francis Group, 2nd Edition, 2023, ISBN: 9781003313007
2.	Hydrogen Fuel: Production, Transport and Storage, Gupta, R. B., CRC Press, Taylor & Francis Group, 1st Edition, 2009, ISBN: 9780429147364
2.	Hydrogen Production: Electrolysis, Agata Godula-Jopek, Wiley-VCH, 1st Edition, 2015, ISBN:9783527333424
3.	Handbook of Hydrogen Storage, Michael Hirscher, Wiley-VCH, 1st Edition, 2010, ISBN:9783527322732
4.	Fuel Cell Systems Explained, James Larminie and Andrew Dicks, John Wiley & Sons, 2nd Edition, 2003, ISBN 978 0470 848579

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	COMPONENTS	MARKS
1.	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. **THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.**	**20**
2.	**TESTS:** Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). THREE tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 150 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) **ADDING UPTO 40 MARKS**.	**40**
	MAXIMUM MARKS FOR THE CIE (THEORY AND PRACTICE)	**100**

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	20
	PART B (Maximum of TWO Sub-divisions only)	
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
	TOTAL	**100**

Semester: VII						
CHEMISTRY OF MATERIALS AND MOLECULAR ANALYSIS **Institutional Elective – II (Group G)** **(Theory)**						
Course Code	**:**	**CM375TGE**		**CIE**	**:**	**100 Marks**
Credits: L:T:P	**:**	**3:0:0**		**SEE**	**:**	**100 Marks**
Total Hours	**:**	**45L**		**SEE Duration**	**:**	**3 Hours**

UNIT – I	**09 Hrs**
Advanced Materials for Smart Packaging: Biodegradable polymers, Bio-polymers, Nanomaterials, and Active packaging materials-Self-healing films, antimicrobial coatings and moisture-sensitive barriers for food and pharmaceutical industries. Thermal-responsive packaging used in food safety, drug stability and electronics protection.	
UNIT – II	**09 Hrs**
Innovations in Computational Chemistry: Introduction to ChemDraw and its Interface- basic to advanced functionalities for chemical drawing, reaction representation, and integration with computational tools. Cheminformatics and molecular screening. - molecular modeling and simulation techniques- molecular docking, virtual screening, and quantitative structure-activity relationships (QSAR)- generative AI for novel drug molecules- neural networks for toxicity prediction and AI-assisted biomolecular structure prediction.	
UNIT – III	**09 Hrs**
IR Spectroscopy: Introduction, principle, molecular vibrations, vibrational frequency, number of fundamental vibrations, factors influencing fundamental vibrations, instrumentation of IR spectrophotometer, application of IR spectroscopy in characterization of functional groups- IR spectroscopy in forensic science, pharmaceuticals, environmental monitoring and material characterization.	
UNIT – IV	**09 Hrs**
UV- visible Spectroscopy: Introduction-Electronic transitions- factors influencing position and intensity of absorption bands-absorption spectra of dienes, polyene and α, β-unsaturated carbonyl compounds, Bandgap calculations utilizing UV. Working of UV-Vis spectrophotometer. AI-driven spectral analysis, and computational approaches for UV-Vis data interpretation. UV-Vis spectroscopy in nanomaterials and biomedical applications.	
UNIT – V	**09 Hrs**
NMR spectroscopy: H^1 NMR Spectroscopy: Basic concepts- relaxation process. NMR spectrometer-FT NMR-Solvents used in NMR, internal standards-chemical equivalence - chemical shift-Factors affecting chemical shifts- shielding and deshielding effects – chemical and magnetic equivalent –magnetic anisotropy-spin-spin splitting rules-Application of NMR on in characterization of compounds- magnetic resonance imaging (MRI)-Bio-Engineering-Bio-Imaging. Problems on prediction of structure of compounds. Basics of Solid State NMR.	

Course Outcomes: After completing the course, the students will be able to	
CO1	Apply the principles of chemistry for the synthesis of smart packaging materials, cheminformatics, drug discovery and materials analysis.
CO2	Utilize the knowledge of chemistry, AI-driven molecular modelling and spectral analysis to identify compounds and predict toxicity.
CO3	Analyse IR, UV-Vis, and NMR spectroscopic data for material characterization and biomolecular studies.
CO4	Propose smart packaging solutions and AI-assisted molecular screening methods for biomedical and pharmaceutical applications

Reference Books	
1.	"Introduction to Spectroscopy" by Donald L. Pavia, Gary M. Lampman, George S. Kriz, and James A. Vyvyan, Published by Cengage Learning, 5th edition, ISBN- 978-1285460123
2.	**"Computational Chemistry: Introduction to the Theory and Applications of Molecular and Quantum Mechanics" –** Errol G. Lewars. Published by Springer 2024, 4th edition, ISBN- 978-3031514425
3.	"Handbook of Materials Structures, Properties, Processing and Performance" by Lawrence E. Murr Published by Springer, ISBN- 978-3319018140.
4.	"Handbook of Biodegradable Polymers-Applications in Biomedical Sciences, Industry, and the Environment" by Shakeel Ahmed and Riyaz Ali Osmani Published by Jenny Standford Publishing 2024, ISBN -9789814968843
5.	"An Introduction to Chemoinformatics" by Andrew R. Leach and Valerie J. Gillet, published by Springer ISBN 978-1402062902.

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
	COMPONENTS	**MARKS**
1.	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO QUIZZES will beconducted & Each Quiz will be evaluated for 10 Marks. **THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.**	**20**
2.	**TESTS:** Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks,adding upto 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specificrequirements (10), Video based seminar/presentation/demonstration (20**) ADDING UPTO 40 MARKS**.	**40**
MAXIMUM MARKS FOR THE CIE THEORY		**100**

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	20
	PART B (Maximum of THREE Sub-divisions only)	
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
	TOTAL	**100**

Semester: VII						
PROMPT ENGINEERING **Institutional Elective – II (Group G)** **(Theory)**						
Course Code	**:**	**CS375TGF**		**CIE**	**:**	**100 Marks**
Credits: L:T:P	**:**	**3:0:0**		**SEE**	**:**	**100 Marks**
Total Hours	**:**	**40 L**		**SEE Duration**	**:**	**03 Hours**

Unit-I	**08 Hrs**
Introduction to Prompt Engineering Raise of Context Learning, Prompts, Prompt Engineering, LLM Settings, Basic Prompt/ Basics of prompting, Elements of a Prompt, Settings for Prompting Language Model/ General Tips for Designing Prompts, Designing Prompts for Different Tasks: few examples of common tasks using different prompts Text Summarization, Information Extraction, Question Answering, Text Classification, Conversation/Role Playing, Code Generation, Reasoning	
Unit – II	**08 Hrs**
Techniques for Effective Prompts Techniques designed to improve performance on complex tasks - Zero-Shot Prompting, Few-shot prompting, Chain-of-thought (CoT) prompting, Zero-Shot CoT, Self-Consistency, Knowledge Generation Prompting, Program-aided Language Model (PAL), ReAct, Directional Stimulus Prompting **Prompt Tuning vs. Fine-Tuning :** Introduction and Types for Tuning, Tools & Libraries for Tuning-Hugging Face PEFT library (Parameter-Efficient Fine-Tuning),Trade-off decisions: when to prompt-tune vs. fine-tune	
Unit –III	**08 Hrs**
Best Practices in Prompt Engineering **Tools & IDEs** Capabilities include: Developing and experimenting with prompts, Evaluating prompts. Versioning and deploying prompts; Advanced prompting techniques : advanced applications with LLMs **LLMs and external tools/APIs** -- LLMs with External Tools; Data-augmented Generation – Steps, External Data, QA with sources, Summarization using sources **Prompt Debugging & Interpretability Tools:** Importance of interpretability in LLMs- Visualization tools- LIT (Language Interpretability Tool by Google) - OpenAI Playground ,debug tools	
Unit –IV	**08 Hrs**
Applications of Prompt Engineering: LLM Applications: Function Calling with LLMs - Getting Started with Function Calling, Function Calling with GPT-4, Function Calling with Open-Source LLMs. **Function Calling Use Cases: Conversational Agents, Natural Language Understanding, Math Problem Solving, API Integration, Information Extraction**	
Unit –V	**08 Hrs**
Ethics in AI-Generated Content: Introduction to AI Ethics in Generation, Key Ethical Concerns in Prompting- Bias and Discrimination, Misinformation and Hallucination. **Opportunities:** Model safety, Prompt Injection, Prompt Leaking, Jail Breaking; Reinforcement Learning from Human Feedback (RLHF)	

Course Outcomes: After completing the course, the students will be able to	
CO1	Demonstrate an understanding of prompt engineering principles including how prompt structure and phrasing impact the performance of AI models.
CO2	Design and implement effective prompts- to create and apply prompts for various natural language processing (NLP) tasks, such as text generation, summarization, and translation, using AI models.
CO3	Critically evaluate the effectiveness of prompts - assess the quality and performance of prompts in terms of accuracy, coherence, and relevance, identifying areas for improvement.
CO4	Apply prompt engineering techniques in real-world scenarios - use prompt engineering strategies to address practical problems in domains such as education, healthcare, and business, demonstrating the applicability of AI-driven solutions.

Reference Books	
1	Unlocking the Secrets of Prompt Engineering, Gilbert Mizrahi, Jan 2024, 1st Edition, Packt Publishing, ISBN-13:978-1835083833
2	Prompt Engineering for Generative AI, James Phoenix, Mike Taylor, May 2024, O'Reilly Media, Inc.,ISBN: 9781098153434
3	Prompt Engineering for LLMs, John Berryman, Albert Ziegler, O'Reilly Media, Inc.January 2025, ISBN: 9781098156152
4	**OpenAI Cookbook-**https://github.com/openai/openai-cookbook **"Ethics of Artificial Intelligence and Robotics"** – Stanford Encyclopedia of Philosophy-https://plato.stanford.edu/entries/ethics-ai/
5	**"The Art of Prompt Engineering with OpenAI API" by Nathan Hunter** (Beginner-friendly, focused on real-world use and debugging)ISBN: 9798389421630

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	**COMPONENTS**	**MARKS**
1.	**QUIZZES:** Quizzes will be conducted in online/offline mode. **TWO QUIZZES** will be conducted & Each Quiz will be evaluated for 10 Marks. **THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.**	**20**
2.	**TESTS:** Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). **TWO tests will be conducted**. Each test will be evaluated for **50 Marks**, adding up to 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (10) Real time problem solving (10) **ADDING UPTO 40 MARKS**.	**40**
	MAXIMUM MARKS FOR THE CIE THEORY	**100**

RUBRIC FOR THE SEMESTER END EXAMINATION (THEORY)		
Q. NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type of questions covering entire syllabus	20
	PART B (Maximum of THREE Sub-divisions only)	
2	Unit 1: (Compulsory)	16
3 & 4	Unit 2: Question 3 or 4	16
5 & 6	Unit 3: Question 5 or 6	16
7 & 8	Unit 4: Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
	TOTAL	**100**

Semester: VII						
SOLID WASTE MANAGEMENT AND STATUTORY RULES **Institutional Elective – II (Group G)** **(Theory)**						
Course Code		**CV375TGG**		**CIE**	**:**	**100 Marks**
Credits: L:T:P	**:**	**3:0:0**		**SEE**	**:**	**100 Marks**
Total Hours	**:**	**42L**		**SEE Duration**	**:**	**3 Hours**

Unit-I	**10 Hrs**
Introduction: Present solid waste disposal methods. Merits and demerits of open dumping, incineration, pyrolysis, composting, sanitary landfill, Biogas Scope and importance of scientific solid waste management. Definition and functional elements of solid waste management. **Sources**: Sources of Solid waste, types of solid waste, composition of municipal solid waste, generation rate, Numerical Problems. **Collection and transportation of municipal solid waste:** Collection of solid waste- services and systems, Municipal Solid waste (Management and Handling) 2016 rules with amendments. Draft SWM 2024 rules. Site visit to collection system.	
Unit – II	**08 Hrs**
Composting Aerobic and anaerobic composting - process description, process microbiology, Vermicomposting, Site visit to compost plant, Numerical problems. **Sanitary land filling**: Definition, advantages and disadvantages, site selection, methods, reaction occurring in landfill- Gas and Leachate movement, Control of gas and leachate movement, Site visit to landfill site.	
Unit –III	**08 Hrs**
Hazardous waste management: Definitions, Identification of hazardous waste, Classification of hazardous waste, onsite storage, collection, transfer and transport, processing, disposal, Hazardous and other wastes (Management and Transboundary Movement) Rules, 2016 with amendments. Hazardous and other wastes (Management and Transboundary Movement) amendment rules, 2024. Site visit to hazardous landfill site	
Unit –IV	**08 Hrs**
Bio medical waste management: Classification of bio medical waste, collection, transportation, disposal of bio medical waste, Biomedical waste management (Management & Handling Rules) 2016 with amendments. Site visit to hospital to observe biomedical waste collection and transportation system and visit to biomedical waste incineration plant.	
Unit –V	**08 Hrs**
E-waste management: Definition, Components, Materials used in manufacturing electronic goads, Recycling and recovery integrated approach. E-waste (Management) Rules 2022 and amendments. Site visit to e- waste treatment plant. **Plastic waste management:** Manufacturing of plastic with norms. Plastic waste management. Plastic waste management rules 2016 with amendments.	

Course Outcomes: After Completing the course, student will be able to,	
CO1	Understand the current solid waste management system and statutory rules.
CO2	Analyze drawbacks in the present system and provide recycling and disposal options for each type of waste in compliance to rules.
CO3	Distinguish Hazardous waste, Biomedical waste, E waste and to provide scientific management system.
CO4	Evaluate and monitor the Biomedical waste, Hazardous waste, E waste, Plastic and Municipal waste management as per the rules laid by Ministry of Environment, Forest and Climate change.

Reference Books	
1.	Integrated Solid Waste Management, George.C. Tchobanoglous, International edition 2022, McGraw hill publication. ISBN 978-0070632370
2.	Electronic waste management , R.E. Hester, Roy M Harrison, , Cambridge, UK, 2009, RSC Publication, ISBN 9780854041121
3.	Solid Waste Management Rules 2016 , Ministry of Environment, Forest and Climate Change Notification, New Delhi, 8th April 2016

4.	Hazardous and other wastes (Management and Transboundary Movement) Rules, 2016, Ministry of Environment, Forest and Climate Change Notification, New Delhi, 04th April, 2016.
5.	Biomedical waste management (Management & Handling Rules) 2016. Ministry of Environment & Forest Notification, New Delhi, amendment on 28th March, 2016.
6.	E-waste (Management) Rules 2022 and amendments , Ministry of Environment, Forest and Climate Change Notification, New Delhi, 2nd November , 2022.
7.	Plastic waste management rules 2016 with amendments., Ministry of Environment, Forest and Climate Change Notification, New Delhi, 6th July 2022.

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	**COMPONENTS**	**MARKS**
1.	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. **THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.**	**20**
2.	**TESTS:** Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). THREE tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 150 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20**) ADDING UPTO 40 MARKS**.	**40**
	MAXIMUM MARKS FOR THE CIE (THEORY AND PRACTICE)	**100**

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	20
	PART B (Maximum of TWO Sub-divisions only)	
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
	TOTAL	**100**

Semester: VII						
FREIGHT TRANSPORTATION SYSTEMS AND LOGISTICS **Institutional Elective – II (Group G)** **(Theory)**						
Course Code	**:**	**CV375TGH**		**CIE**	**:**	**100 Marks**
Credits: L:T:P	**:**	**3:0:0**		**SEE**	**:**	**100 Marks**
Total Hours	**:**	**42L**		**SEE Duration**	**:**	**3 Hours**

Unit-I	**08 Hrs**
Characteristics of Freight Transport: Freight Characteristics, Factors influencing Freight Travel, operators, problems in freight transportation, regional and urban goods travel, intermodal freight travel issues, passenger and freight demand models.	
Unit – II	**09 Hrs**
Freight Demand Estimation: Operations, Planning - purpose, process, Data, Freight Agents, costs, Planning Models and Methods-freight demand estimation and forecasting at the regional and urban level, Freight Generation and Freight Trip Generation, Trend and time series models, freight trip rate models, IO model	
Unit –III	**08 Hrs**
Freight Transport Planning and Operations: Freight supply – capacity issues; freight productivity and performance; distribution of freight flows; production/consumption to origin/destination, competing modes for specific commodity choice, route planning, scheduling, collection storage, distribution centers, regulation, and enforcement of freight transport.	
Unit –IV	**09 Hrs**
Logistics and Planning Strategies: Context of Logistics- Activities of Logistics, Aims of Logistics, Importance of Logistics, Current Trends in Logistics; Logistics Strategy- Strategic Decisions, Logistics Strategy, Designing a Logistics Strategy; Locating Facilities- Importance of Location, Choosing the Geographic Region, Infinite Set Approaches, Feasible Set Approaches, Network Models, Location Planning; Planning Resources- Types of Planning, Capacity Planning, Adjusting Capacity, Tactical Planning, Schedules	
Unit –V	**08 Hrs**
Emerging Trends and Case Studies in Transportation Logistics Sustainable and Green Logistics, **Digital Transformation in Logistics**-IoT, big data analytics, AI, and blockchain in logistics management. **Case Studies and Real-World Applications** Case studies on logistics and transportation from various sectors. Analysis of successful logistics systems and lessons learned.	

Course Outcomes	
CO1	Explain the characteristics, components, and current trends in freight transportation and logistics
CO2	Analyze and apply freight demand estimation methods and logistics planning strategies for urban and regional transport systems
CO3	Evaluate freight transport systems and logistics networks with respect to performance, capacity, sustainability, and regulation
CO4	Design integrated freight and logistics solutions using data-driven models, technology, and strategic planning approaches

Reference Books	
2.	M. Ben-Akiva, H. Meersman, and E. V. de Voorde, "*Freight Transport Modelling*" Bingley, U.K.: Emerald Group Publishing, **ISBN-13:** 9781781902851,2013.
2.	P. K. Sarkar, V. Maitri, and G. J. Joshi, "*Transportation Planning: Principles, Practices and Policies"*, 3rd ed. New Delhi, India: PHI Learning, ISBN : 9788195161188, 2024
3.	L. Tavasszy and G. de Jong, "*Modelling Freight Transport"*. 1st ed. Amsterdam, Netherlands: Elsevier, **ISBN-13:** 9780124104006,2013.
4	M. Al-Azzawi, "*Freight and Logistics Transport Modelling and Planning: Mathematical Theories and Practical Applications for the Analysis and Forecasting of Freight Transport Systems"*. Saarbrücken, Germany: LAP Lambert Academic Publishing, **ISBN-13:** 9783848428380,2012

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	COMPONENTS	MARKS
1.	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. **THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.**	**20**
2.	**TESTS:** Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). THREE tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 150 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) **ADDING UPTO 40 MARKS**.	**40**
	MAXIMUM MARKS FOR THE CIE (THEORY AND PRACTICE)	**100**

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	20
	PART B (Maximum of TWO Sub-divisions only)	
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
	TOTAL	**100**

Semester: VII						
IoT FOR SMART SYSTEMS **Institutional Elective – II (Group G)** **(Theory)**						
Course Code	**:**	**EC375TGI**		**CIE**	**:**	**100 Marks**
Credits: L:T:P	**:**	**3:0:0**		**SEE**	**:**	**100 Marks**
Total Hours	**:**	**45L**		**SEE Duration**	**:**	**3 Hours**

Unit-I	**09 Hrs**
Introduction to IoT and Networking Basics IoT Architecture Layers (Perception, Network, Application), Overview of Networking Basics: TCP/IP, OSI Model, Wireless Concepts, Device Addressing: IPv4, IPv6, 6LoWPAN, IoT orchestration primitives, IoT Open Source platforms.	
Unit – II	**09 Hrs**
IoT Communication Models and Protocols **Smart Manufacturing**-OPC-UA interfacing, Deterministic Ethernet & Time Sensitive Network, PLC gateways, MQTT-Sparkplug B, Predictive maintenance dashboard design with AI-based Optimization. **Smart Construction Site:** MQTT with QoS for equipment-telemetry and asset tracking, Edge gateway (rugged fog node) with aggregating sensor feeds (On-site construction).	
Unit –III	**09 Hrs**
IoT Networking Technologies **Smart Healthcare**: IEEE 802.15.6 BLE Body Area Networks, NFC for patient data access and device pairing, HL7 and FHIR standards for health data exchange. **Smart Mobility**: Next-gen V2X, LTE-M for roadside units and environmental sensors, Edge-cloud split for traffic-signal optimisation, LPWAN & NB-IoT with link-budget exercise.	
Unit –IV	**09 Hrs**
IoT Software Stack **Smart Agriculture Systems:** Device Layer: Sensor nodes run on MCUs with firmware enabling periodic sensing, sleep scheduling, and uplink messaging over LPWAN. Network Layer: Embedded Operating Systems for IoT (FreeRTOS). Middleware Layer: Gateway running rule engine and device management, Device abstraction and Service Enablement, Secure bootloaders for wireless firmware updates in remote farms. Application Layer: Web/mobile dashboard for farmers with real-time alerts and analytics, ThingsBoard for drag-and-drop rule engine.	
Unit –V	**09 Hrs**
Interoperability and Standardization in Practice: Challenges of integrating multi-vendor IoT devices, Interoperability issues in AI-Enabled IoT Deployments, Data Semantics & Model Portability, AI Bias & Fairness Across Domains, Security of AI pipelines, GDPR compliance & data retention policies in IoT platforms**.**	

Course Outcomes	
CO1	Develop a conceptual framework of IoT architectural layers and connectivity models.
CO2	Analyze IoT-specific communication models, protocols, and networking technologies.
CO3	Apply AI techniques for IoT communication optimization and network intelligence.
CO4	Evaluate secure, energy-efficient IoT networks with intelligent features.

Reference Books	
1.	Internet of Things: Principles and Paradigms, Rajkumar Buyya, Amir Vahid Dastjerdi, Morgan Kaufmann Publishers, 2nd Edition, 2024, ISBN: 978-0-323-99167-2
2.	Architecting the Internet of Things, Dieter Uckelmann, Mark Harrison, Florian Michahelles Springer, 2nd Edition, 2023 ISBN: 978-3-031-10318-4
3.	IoT Applications for Next-Generation Smart Systems, D. Giusto et al., Springer,2024,ISBN:978-1-799-87541-3
4.	Artificial Intelligence for the Internet of Everything, William Lawless, Ranjeev Mittu, Donald Sofge Academic Press, 2023 ISBN: 978-0-12-820600-1

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	COMPONENTS	MARKS
1.	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. **THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.**	**20**
2.	**TESTS:** Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). THREE tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 150 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) **ADDING UPTO 40 MARKS**.	**40**
	MAXIMUM MARKS FOR THE CIE (THEORY AND PRACTICE)	**100**

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	20
	PART B (Maximum of TWO Sub-divisions only)	
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
	TOTAL	**100**

Semester: VII						
E-MOBILITY **Institutional Elective – II (Group G) (Theory)**						
Course Code	**:**	**EE375TGJ**		**CIE**	**:**	**100 Marks**
Credits: L:T:P	**:**	**3:0:0**		**SEE**	**:**	**100 Marks**
Total Hours	**:**	**45L**		**SEE Duration**	**:**	**3 Hours**

Unit-I	**09 Hrs**
History, Basics of Electric Vehicles, Components of Electric Vehicle, General Layout of EV, EV classification: Battery Electric Vehicles (BEVs), Hybrid Electric Vehicle (HEV), Fuel-Cell Electric Vehicles (FCEVs) Comparison with Internal Combustion Engine: Technology, Advantages & Disadvantages of EV, National Policy for adoption of EVs.	
Unit – II	**09 Hrs**
Electric Drive-Trains: Introduction to various electric drive-train topologies in EV and HEV, Power flow control in electric drive-train topologies, classification of electric machines used in automobile drivetrains. **E-Motor Drives Configuration (Control Block diagrams):** Induction Motor Drive, Permanent Magnet (PM) motor Drive & Switched Reluctance Motor (SRM) Drive.	
Unit –III	**09 Hrs**
Battery Energy Storage: Types of Battery, Introduction to Electrochemical Battery, Electrochemical Reactions, Battery Parameters: Battery Capacity, Discharge Rate, Charging Rate, SOC, SOD, SOH, DOD, Specific Energy, Specific Power, Energy Efficiency. **Battery Management Systems (BMS):** Introduction to BMS, Objectives of the BMS: Discharging control, Charging control, Cell Balancing; BMS topologies: Distributed Topology, Modular Topology and Centralized Topology.	
Unit –IV	**09 Hrs**
Energy Storage: Introduction to Energy Storage Requirements in Hybrid and Electric Vehicles, Battery based energy storage, Fuel Cell based energy storage, Super Capacitor based energy storage, Hybridization of different energy storage devices. Introduction to BMS and its topologies. **Energy Management Strategies**: Introduction to energy management strategies used in hybrid and electric vehicles, Classification of different energy management strategies, Comparison of different energy management strategies and implementation issues of energy management strategies.	
Unit –V	**09 Hrs**
Charging Infrastructure: Basic Requirements for Charging System, Charger Architectures, Grid Voltages, Frequencies, and Wiring, Charging Standards and Technologies, SAE J1772. On-board chargers and Off-board chargers, Topologies and Standards, Types of Charging Station Charging Station Placement for Electric Vehicles: A Case Study.	

Course Outcomes	
CO1	Analyse the basics of electric and hybrid electric vehicles, their architecture, technologies and modelling.
CO2	Analyse various electric drives suitable for electric vehicles.
CO3	Discuss and implement different energy storage technologies used for electric vehicles and their management system.
CO4	Analyse various charging methods, requirements, standards and types of charging for EV and HEV.

Reference Books	
1.	Modern Electric Vehicle Technology C.C. Chan and K.T. Chau, 1st Edition, 2001, Oxford university press, ISBN 0 19 850416 0.
2.	Battery Management system for large Lithium Battery Packs, Davide Andrea, 1st Edition, 2010, ARTECH HOUSE, ISBN-13 978-1-60807-104-3.
3.	Electric Powertrain: Energy Systems, Power Electronics and Drives for Hybrid, Electric and Fuel Cell Vehicles, John G. Hayes, G. Abas Goodarzi, 1st Edition, 2018, Wiley, ISBN 9781119063667.
4.	Hybrid Vehicles from Components to System, F. BADIN, Ed, 1st Edition, 2013, Editions Technip, Paris, ISBN 978-2-7108-0994-4.

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	COMPONENTS	MARKS
1.	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. **THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.**	**20**
2.	**TESTS:** Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). THREE tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 150 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20**)** **ADDING UPTO 40 MARKS**.	**40**
	MAXIMUM MARKS FOR THE CIE (THEORY AND PRACTICE)	**100**

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	20
	PART B (Maximum of TWO Sub-divisions only)	
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
	TOTAL	**100**

Semester: VII						
DISEASE AND DIAGNOSTICS AN ENGINEERING PERSPECTIVE **Institutional Electives -II (Group G)** **(Theory)**						
Course Code	**:**	**EI375TGK**		**CIE**	**:**	**100 Marks**
Credits: L:T:P	**:**	**3:0:0**		**SEE**	**:**	**100 Marks**
Total Hours	**:**	**45L**		**SEE Duration**	**:**	**03 Hrs**

Unit-I	**09 Hrs**
Cardiovascular System(CVS) - Functioning of CVS, Generation of Bio-Potentials of CVS, Common cardiovascular ailments. - Arrhythmias, Hypertension, Coronary vascular Diseases, Diagnostic devices- ECG, ECHO. Coronary Angiogram. Implantable devices: Pacemaker.	
Unit – II	**09 Hrs**
Respiratory System(RS) - Functioning of RS, Common ailments in RS, Lung volume and capacity, Diagnostic Devices- Spirometer and its recent advances, PPG Signals and its application in RS.	
Unit –III	**09 Hrs**
Renal System - Functioning of Renal System, Common diseases of Renal System, Diagnostics- Ultrasound and its application in Renal System. Therapeutic- Dialysis and its types. Bio-Markers in Renal System, Emerging dialysis technologies: Wearable artificial kidney.	
Unit –IV	**09 Hrs**
Central Nervous System(CNS) - Basics of CNS functioning, Common ailments, Diagnostics methods-EEG, Electrode Systems, Role of CT Scans and MRI. Brain-Computer Interfaces (BCI): Basic concepts and diagnostic/therapeutic potential.	
Unit –V	**09 Hrs**
Introduction to Artificial Intelligence(AI) in Healthcare - Overview of AI, Machine Learning, Deep Learning, Need for AI in healthcare, Types of data in healthcare: structured, unstructured., Challenges of AI deployment: Data bias, privacy, and integration in clinical workflow.	

Course Outcomes	
CO1	Understand the basic functioning and common disorders of cardiovascular, respiratory, renal, and nervous systems.
CO2	Understand the working principles and clinical relevance of diagnostic devices such as ECG, EEG, spirometer, ultrasound, and imaging systems.
CO3	Analyze biomedical signals and interpret their significance in the diagnosis and monitoring of diseases.
CO4	Evaluate the potential and limitations of emerging technologies such as wearable devices, Brain-Computer Interfaces, and AI in healthcare applications.

Reference Books	
1.	Handbook of Biomedical Instrumentation, R. S. Khandpur,3rd Edition, Reprint 2016, Tata McGraw-Hill, ISBN: 9780070473553.
2.	Biomedical Instrumentation and Measurements, Leslie Cromwell & others, 2nd Edition, Reprint 2015, ISBN: 9780130771315.
3.	Introduction to Biomedical Equipment Technology, Joseph J. Carr and John M. Brown, 4th Edition, 2000, Pearson, ISBN:978-0130104922**.**
4.	Medical Instrumentation Application and Design, John G. Webster ,4th Edition,2010, ISBN 13: 978- 0471-67600-3.

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	**COMPONENTS**	**MARKS**
1.	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. **THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.**	**20**
2.	**TESTS:** Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). THREE tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 150 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) **ADDING UPTO 40 MARKS**.	**40**
	MAXIMUM MARKS FOR THE CIE (THEORY)	**100**

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	20
	PART B (Maximum of TWO Sub-divisions only)	
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
	TOTAL	**100**

Semester: VII						
SPACE TECHNOLOGY AND APPLICATIONS **Institutional Elective Course** **(Theory)**						
Course Code	**:**	**ET375TGL**		**CIE**	**:**	**100 Marks**
Credits: L:T:P	**:**	**3:0:0**		**SEE**	**:**	**100 Marks**
Total Hours	**:**	**45L**		**SEE Duration**	**:**	**03 Hours**

UNIT – I	**09 Hrs**
Orbits and Launching Methods: Kepler's Laws, Orbital Elements and Perturbations, Inclined Orbits Antenna Look Angles, Earth Eclipse of Satellite, Sun Transit Outage. **Launch Vehicles:** Rocketry, Propellants, Propulsion, Combustion, Solid, Liquid and Cryogenic engines, Control and Guidance system, Ion propulsion and Nuclear Propulsion.	
UNIT – II	**09 Hrs**
Satellite Sub-Systems: Classification of satellites., Payloads, Subsystems-Altitude and orbit control system, TT&C Sub-System, Power Systems. **Communication sub systems:** Transponder, Satellite antennas and Parameters, Satellite Antennas in Practice	
UNIT – III	**09 Hrs**
Satellite Communications: LEO, MEO and GEO orbits, Multiple Access Techniques. -TDMA, FDMA and CDMA **Space applications:** Telephony, V-SAT, DBS system, Satellite Radio and TV, Tele-Education, Tele-medicine, Satellite navigation, GPS	
UNIT – IV	**09 Hrs**
Remote Sensing: Remote Sensing – An Overview, Classification of Satellite Remote Sensing Systems, Remote Sensing Satellite Payloads, Sensor Parameters. Image processing techniques, Remote sensing cameras. **Applications of Remote Sensing Satellites**: - Land Cover Classification, Land Cover Change Detection Water Quality Monitoring and Management, Flood Monitoring, weather modelling, Cyclone predictions, Disaster and flood warning, rainfall predictions using satellites.	
UNIT – V	**09 Hrs**
Space Missions: Technology missions, deep space planetary missions, Lunar missions, zero gravity experiments, space biology and international space Missions. **Advanced space systems:** planetary payloads, space shuttle, space station, Inter-space communication systems.	

Course Outcomes: After going through this course the student will be able to:	
CO1	Explore the different types of satellites, their orbital mechanics and associated subsystems.
CO2	Apply the basics of launching vehicles, satellites and sub systems for space applications.
CO3	Identify the applications of satellite in the Various areas such as communication, remote sensing, metrology etc.
CO4	Analyze the technology trends, satellite missions and advanced space systems.

Reference Books
1.Satellite Communications, Dennis Roddy, W. Linwood Jones, Jones Linwood, David G. Long, 5ed, McGraw Hill Education,2024, ISBN-10: 1265372543
2. Fundamentals of Satellite Communication, K N Raja Rao, PHI, 2012, ISBN: 9788120324015.
3. Satellite Communication, Timothy Pratt, John Wiley, 3ed,2019, ISBN: 978-1-119-48217-8
4. Remote sensing and applications, B C Panda, VIVA books Pvt. Ltd., 2009, ISBN: 108176496308.

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (CIE-Theory)		
CIE will consist of TWO Quizzes (Q), TWO Tests (T), and ONE Experiential Learning (EL) component **[20 (Q) + 40 (T) + 40 (EL) = 100 marks)**		
Sl.No.	**COMPONENTS**	**MARKS**
1.	**QUIZZES:** Quizzes will be conducted in online/offline mode. **TWO QUIZZES** will be conducted & each Quiz will be evaluated for 10 marks, and Final Quiz marks adding up to 20 marks. **THE SUM OF TWO QUIZZES WILL BE CONSIDERED AS FINAL QUIZ MARKS.**	**20**
2.	**TESTS:** Students will be evaluated in test consisting of descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). **TWO TESTS** will be conducted. Each test will be evaluated for 50 Marks, adding up to 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. **Phase I (20) & Phase II (20) ADDING UPTO 40 MARKS**.	**40**
	MAXIMUM MARKS FOR THE CIE	**100**

RUBRIC FOR SEMESTER END EXAMINATION (SEE-Theory)		
Q.NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type of questions covering entire syllabus	20
	PART B (Maximum of THREE Sub-divisions only)	
	Unit 1: Question 2	16
	Unit 2: Question 3 or 4	16
	Unit 3: Question 5 or 6	16
	Unit 4: Question 7 or 8	16
	Unit 5: Question 9 or 10	16
	MAXIMUM MARKS FOR THE SEE	**100**

Semester: VII						
PROJECT MANAGEMENT **Institutional Elective – II (Group G)** **(Theory)**						
Course Code	**:**	**IM375TGM**		**CIE**	**:**	**100 Marks**
Credits: L:T:P	**:**	**3:0:0**		**SEE**	**:**	**100 Marks**
Total Hours	**:**	**45L**		**SEE Duration**	**:**	**3 Hours**

Unit-I	**09 Hrs**
Introduction: Project, Project management, relationships among portfolio management, program management, project management, and organizational project management, relationship between project management, operations management and organizational strategy, business value, role of the project manager, project management body of knowledge. **Generation and Screening of Project Ideas:** Generation of ideas, monitoring the environment, corporate appraisal, scouting for project ideas, preliminary screening, project rating index, sources of positive net present value.	
Unit – II	**09 Hrs**
Project Scope Management: Project scope management, collect requirements define scope, create Work Breakdown Structure, validate scope, control scope. **Organizational influences & Project life cycle**: Organizational influences on project management, project state holders & governance, project team, project life cycle.	
Unit –III	**09 Hrs**
Project Integration Management: Develop project charter, develop project management plan, direct & manage project work, monitor & control project work, perform integrated change control, close project or phase. **Project Quality management:** Plan quality management, perform quality assurance, control quality.	
Unit –IV	**09 Hrs**
Project Risk Management: Plan risk management, identify risks, perform qualitative risk analysis, perform quantitative risk analysis, plan risk resources, control risk. **Project Scheduling:** Project implementation scheduling, Effective time management, Different scheduling techniques, Resources allocation method, Project life cycle management concepts. Project life cycle costing.	
Unit –V	**09 Hrs**
Latest Trends and Open-Source Tools in Project Management: Agile and Hybrid Project Management Approaches, Digital Transformation in Project Management, Sustainability and ESG in Project Management, Overview and Classification of PM Software, Introduction and Demonstration of Key Open-Source Tools.	

Course Outcomes	
CO1	Understand the fundamental concepts of project management and its relationship with organizational strategy, operations management, and business value.
CO2	Apply techniques for generating, screening, and evaluating project ideas, considering factors such as net present value and project rating index.
CO3	Create Work Breakdown Structures (WBS), utilization of PERT/CPM for developing project schedule, alongside requirement collection, scope definition, scope validation, and scope control.
CO4	Develop skills in project integration, quality, risk management, and scheduling, enabling effective project planning, execution, monitoring, and control.

Reference Books	
1.	Project Management Institute, "A Guide to the Project Management Body of Knowledge (PMBOK Guide)", 5th Edition, 2013, ISBN: 978-1-935589-67-9
2.	Harold Kerzner, Project Management A System approach to Planning Scheduling & Controlling, John Wiley & Sons Inc., 11th Edition, 2013, ISBN 978-1-118-02227-6.
3.	Prasanna Chandra, Project Planning Analysis Selection Financing Implementation & Review, Tata McGraw Hill Publication, 7th Edition, 2010, ISBN 0-07-007793-2.
4.	Rory Burke, "Project Management – Planning and Controlling Techniques", John Wiley & Sons, 4th Edition, 2004, ISBN: 9812-53-121-1

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	COMPONENTS	MARKS
1.	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. **THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.**	**20**
2.	**TESTS:** Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). THREE tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 150 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20**) ADDING UPTO 40 MARKS**.	**40**
	MAXIMUM MARKS FOR THE CIE (THEORY AND PRACTICE)	**100**

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	20
	PART B (Maximum of TWO Sub-divisions only)	
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
	TOTAL	**100**

Semester: VII						
GLOBAL SUPPLY CHAIN MANAGEMENT **Institutional Elective – II (Group G)** **(Theory)**						
Course Code	**:**	**IM375TGN**		**CIE**	**:**	**100 Marks**
Credits: L:T:P	**:**	**3:0:0**		**SEE**	**:**	**100 Marks**
Total Hours	**:**	**45L**		**SEE Duration**	**:**	**3 Hours**

UNIT-I	**09 Hrs**
Building a Strategic Frame Work to Analyse Supply Chains: Definition and Objective of Supply Chain, the importance of Supply Chain Decisions, Decision Phases in a Supply Chain, Process View of Supply Chains. Competitive and Supply Chain Strategies, Achieving Strategic fit, Expanding Strategic Scope. Drivers of Supply Chain Performance, Frame work for Structuring Drivers, Facilities, Inventory, Transportation, Information, Sourcing, Pricing, Infrastructure, International Logistics.	
UNIT – II	**09Hrs**
Designing The Supply Chain Network: The Role of Distribution in the Supply Chains, Factors influencing Distribution Network design, Design Options for a Distribution Network, Online sales and the Distribution network, Distribution Networks in practice. Factors influencing network design decisions, Framework for Network design decisions, The impact of Globalization on Supply Chain networks. **Designing And Planning Transportation Networks:** The role of transportation in a Supply chain, Modes of transportation and their performance characteristics, Transportation infrastructure and policies, Design options for a transportation network, Trade-offs in transportation design, Tailored transportation, The role of IT in transportation, **Problems.**	
UNIT –III	**09Hrs**
Planning and Managing Inventories in a Supply Chain: The Role of Cycle inventory in a Supply Chain, Economies of Scale to Exploit Fixed costs, Managing Multi-echelon Cycle Inventory. The Role of Safety Inventory in a Supply Chain, Determining appropriate level of Safety inventory, Impact of supply Uncertainty on Safety inventory, Impact of aggregation on safety inventory, impact of replenishment policies on safety inventory, Managing Safety Inventory in a Multi-echelon Supply Chain, The Role of IT in inventory management. **Problems**	
Unit –IV	**09Hrs**
Sourcing Decisions in A Supply Chain: The role of sourcing in a supply chain, in-house or outsource, Third-and Fourth-party logistics providers, Total cost of Ownership, Supplier Selection-Auctions and Negotiations, Sharing Risk and Reward in the Supply chain, Ethical Sourcing. **Pricing and Revenue management in Supply Chain**: The role of pricing and revenue management in supply chain, pricing and revenue management for multiple customer segments, pricing and revenue management for perishable assets, pricing and revenue management for seasonal demand and role of IT in pricing and revenue management.	
UNIT –V	**09 Hrs**
Digital Supply Chain: The role of IT in a supply chain, the supply chain IT framework, the supply chain macro processes, Lack of Supply Chain co-ordination and the Bullwhip effect, managerial levers to achieve coordination, continuous replenishment and vendor-managed inventories, collaborative planning, forecasting and replenishment (CPFR).	

Course Outcomes	
CO1	Understand supply chain concepts, systemic and strategic role of SCM in global competitive environment.
CO2	Evaluate alternative supply and distribution network structures using optimization models.
CO3	Develop optimal sourcing and inventory policies in the supply chain context.
CO4	Select appropriate information technology frameworks for managing supply chain processes.

Reference Books	
1.	Supply Chain Management – Strategy, Planning & Operation, Sunil Chopra, Peter Meindl & D V Kalra, 6th Edition, 2016, Pearson Education Asia; ISBN: 978-0-13-274395-2.
2.	Supply Chain Management – Creating Linkages for Faster Business Turnaround, Sarika Kulkarni & Ashok Sharma, 1st Edition, 2004, TATA Mc Graw Hill, ISBN: 0-07-058135–5
3.	Designing & Managing the Supply Chain – Concepts Strategies and Case Studies, David Simchi Levi, Philip Kaminsky, Edith Simchi Levi & Ravi Shankar, 3rd Edition, 2008, Mc Graw Hill, ISBN: 978- 0-07-066698-6
4.	Modelling the Supply Chain, Jeremy F Shapiro, 2nd Edition, 2009, Cengage Learning, ISBN 0-495-12609-8.

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	**COMPONENTS**	**MARKS**
1.	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. **THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.**	**20**
2.	**TESTS:** Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). THREE tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 150 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) **ADDING UPTO 40 MARKS**.	**40**
	MAXIMUM MARKS FOR THE CIE (THEORY AND PRACTICE)	**100**

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	20
	PART B (Maximum of TWO Sub-divisions only)	
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
	TOTAL	**100**

Semester: VII						
STATISTICAL METHODS FOR ENGINEERS **Institutional Elective – II (Group G)** **(Theory)**						
Course Code	**:**	**MA375TGO**		**CIE**	**:**	**100 Marks**
Credits: L: T:P	**:**	**3:0:0**		**SEE**	**:**	**100 Marks**
Total Hours	**:**	**45L**		**SEE Duration**	**:**	**3.00 Hours**

Unit-I	**09 Hrs**
Multiple Linear Regression: Introduction to multiple linear regression model, Least squares estimation of the parameters, Matrix approach to multiple linear regression, Properties of the least squares estimators, Hypothesis tests -Test for significance of regression, Tests on individual regression coefficients, Confidence intervals - Confidence intervals on individual regression coefficients, Confidence interval on the mean response, Polynomial regression models.	
Unit – II	**09 Hrs**
Design and Analysis of Single-Factor Experiments: Designing Engineering Experiments, Completely Randomized Single-Factor Experiment, Analysis of Variance (ANOVA), Multiple Comparisons following the ANOVA, Residual analysis and model checking, The random-effects model, Randomized complete block design - design and statistical Analysis **Design of Experiments with Several Factors:** Introduction, Factorial experiments, Two-factor factorial experiments, one observation per cell, General factorial experiments - 2^k Factorial designs, 2^2 Design, 2^k Design for $k \geq 3$ factors.	
Unit –III	**09 Hrs**
Discriminant Analysis and Factor Analysis: Introduction, Linear Discriminant Model, Linear discriminant function, Discriminant analysis, Principal Component, Factor Analysis, Principal Components versus Factor Analysis, Applications. **Conjoint Analysis:** Introduction, Additive tables, Multiplicative tables, Computing table margins based on an additive model, Applied conjoint analysis.	
Unit –IV	**09 Hrs**
Decision Trees and Clustering: Introduction, Basic tree model, Categorical or Quantitative Predictors, Decision Trees, Regression Trees, Bias-Variance Tradeoff, Pruning and Cross-Validation, Loss functions**.** Cluster Analysis-Introduction, Types of Clustering, Correlations and Distances, Hierarchical Clustering, Partitioning via k-means.	
Unit –V	**09 Hrs**
Time Series Analysis: Time series analysis-Trend, Seasonality, Noise decomposition, Autocorrelation, Stationarity, Lag features, Overview of Autoregressive (AR), Moving Average (MA), Autoregressive Integrated Moving Average (ARIMA) models.	

Course Outcomes:	
CO1	Apply fundamental concepts of multiple regression, design of experiments to analyze and interpret data in discriminant analysis, factor analysis, conjoint analysis, and statistical learning methods.
CO2	Analyze and interpret data using regression, experimental design, and multivariate techniques, and apply suitable statistical learning models to real-world problems.
CO3	Estimate model parameters and interpret results for multiple linear regression, ANOVA, discriminant analysis, factor analysis, and conjoint analysis to support data-driven decision-making using statistical learning techniques.
CO4	Implement multiple linear regression, ANOVA, discriminant analysis, factor analysis, conjoint analysis and time series analysis to support decision-making using statistical methods for solving engineering problems.

Reference Books	
1	An Introduction to Multivariate Analysis, T. W. Anderson, 3rd Edition, 2003, John Wiley & Sons, New Jersey, ISBN: 0-471-36091-0.
2	Douglas C. Montgomery and George C. Runger, "Applied Statistics and Probability for Engineers", John Wiley & Sons, 7th Edition, 2019, ISBN: 9781119570615.
3	Trevor Hastie, Robert Tibshirani, Jerome Friedman, "The Elements of Statistical Learning: Data Mining, Inference, and Prediction", Springer Series in Statistics, 2nd Edition, 2009.
4	Richard A Jhonson and Dean W. Wichern, Applied Multivariate Statistical Analysis, 6th Edition, Pearson Education, ISBN-13: 978-0-13-187715-3.
5	Statistics I, SYSTAT 10.2, ISBN 81-88341-04-5.

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
	COMPONENTS	**MARKS**
1.	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. **THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.**	**20**
2.	**TESTS:** Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). **TWO tests will be conducted.** Each test will be evaluated for 50 Marks, adding upto 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20**) ADDING UPTO 40 MARKS**.	**40**
	MAXIMUM MARKS FOR THE CIE THEORY	**100**

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	20
	PART B (Maximum of TWO Sub-divisions only)	
2	Unit 1: (Compulsory)	16
3 & 4	Unit 2: Question 3 or 4	16
5 & 6	Unit 3: Question 5 or 6	16
7 & 8	Unit 4: Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
	TOTAL	**100**

Semester: VII						
INDUSTRY 5.0 **Institutional Elective – II (Group G)** **(Theory)**						
Course Code	**:**	**ME375TGP**		**CIE**	**:**	**100 Marks**
Credits: L:T:P	**:**	**3:0:0**		**SEE**	**:**	**100 Marks**
Total Hours	**:**	**42 Hrs**		**SEE Duration**	**:**	**03 Hours**

Unit – I	**06 Hrs**
Evolution to Industry 5.0: Historical overview of industrial revolutions. Transition from automation-centric to human-centric manufacturing. Key drivers: personalization, sustainability, and resilience. Comparison between Industry 4.0 and Industry 5.0 frameworks.	
Unit -II	**08 Hrs**
Human-Machine Collaboration: Introduction to collaborative robots (cobots) and their applications. Designing ergonomic and safe human-robot workspaces. Augmented Reality (AR) and Virtual Reality (VR) in training and maintenance. Case studies on successful human-robot collaboration.	
Unit –III	**08 Hrs**
Advanced Digital Technologies: Digital twins and their role in predictive maintenance and system optimization. Integration of AI and machine learning in manufacturing processes. Cyber-Physical Systems (CPS) and their applications. Edge computing and real-time data analytics.	
Unit –IV	**10 Hrs**
Sustainable and Resilient Manufacturing: Principles of sustainable manufacturing and circular economy. Life Cycle Assessment (LCA) and its importance in product design. Strategies for building resilient supply chains. Regulatory frameworks and standards for sustainable practices.	
Unit - V	**10 Hrs**
Implementation and Case Studies: Roadmap for transitioning to Industry 5.0. Challenges and solutions in adopting Industry 5.0 principles. Case studies from automotive, aerospace, and electronics industries. Future trends and research directions in Industry 5.0.	

Course Outcomes: After completing the course, the students will be able to	
CO1	Assess the challenges of implementing Industry 5.0 principles in diverse industrial applications.
CO2	Analyze the strategies for transitioning from Industry 4.0 to Industry 5.0 in human-centric manufacturing systems.
CO3	Evaluate the benefits of integrating collaborative technologies and sustainable practices in smart manufacturing.
CO4	Examine the role of human-machine collaboration and ethical data-driven decision-making in achieving resilient and adaptive production systems.

Reference Books	
1.	Industry 5.0: The Future of the Industrial Economy, Uthayan Elangovan, 2022, CRC Press, 1st Edition, ISBN: 9781032041278 (Unit I)
2.	Industry 5.0: Concepts and Strategies for Digital Transformation, Kaswan, Rathi, Garza-Reyes, 2025, CRC Press, 1st Edition, ISBN: 9781032878218 (Unit II, V)
3.	Industry 5.0: Design, Standards, Techniques and Applications for Manufacturing, Leong, 2024, The IET, 1st Edition, ISBN: 9781837240098 (Unit IV)
4.	Towards Industry 5.0, Durakbasa, Gencyilmaz, 2022, Springer, 1st Edition, ISBN: 9783031244568 (Unit III, V)
5.	Intelligent Manufacturing: Smart Choice, Sunil Kumar Wadhwa, 2023, ISBN: 9781802279153

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	COMPONENTS	MARKS
1.	**QUIZZES:** Quizzes will be conducted in online/offline mode. **TWO QUIZZES** will be conducted & Each Quiz will be evaluated for 10 Marks. Each quiz is evaluated for 10 marks adding up to 20 MARKS	**20**
2.	**TESTS:** Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) ADDING UPTO 40 MARKS.	**40**
	MAXIMUM MARKS FOR THE CIE THEORY	**100**

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type of questions covering entire syllabus	20
	PART B (Maximum of THREE Sub-divisions only)	
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
	TOTAL	**100**

Semester: VII						
PRINCIPLES OF ASTROPHYSICS **Institutional Elective – II (Group G)** **(Theory)**						
Course Code	**:**	**PY375TGQ**		**CIE**	**:**	**100 Marks**
Credits: L:T:P	**:**	**3:0:0**		**SEE**	**:**	**100 Marks**
Total Hours	**:**	**42L**		**SEE Duration**	**:**	**3 Hours**

Unit-I	**08 Hrs**
The Celestial Sphere: The Geocentric Universe, Retrograde motion, The Copernican revolution, The celestial sphere, Altitude–Azimuth and Equatorial Coordinate System, Precession. **Celestial Mechanics:** Kepler's Laws of Planetary Motion from Newton's laws of mechanics, Position in the Orbit. **Spectrum of Light:** Spectral lines - Kirchhoff's Laws, Blackbody radiation and temperature, Spectrographs, Stellar parallax, The magnitude scale, Apparent and Absolute Magnitude.	
Unit – II	**09 Hrs**
Telescopes: Refracting and Reflecting Telescopes, Resolution and the Rayleigh Criterion, Aberrations, The Brightness of an Image and focal ratio, Adaptive Optics, Light Detectors - Photographic plates, CCD, Radio Telescopes - Spectral Flux Density, Large Apertures and Interferometry, Infrared, Ultraviolet, X-Ray, and Gamma-Ray Astronomy.	
Unit –III	**08 Hrs**
Special Relativity: Einstein's postulates, The relativity of simultaneity, Time dilation, Length contraction, Galilean transformations, Lorentz transformations, Velocity transformation, Space-time diagrams and causality. Relativistic momentum and energy. **General Relativity:** The Principle of Equivalence, Gravitation as space-time curvature, (1) perihelion advance of planet Mercury, (2) gravitational red shift and (3) gravitational bending of light (qualitative).	
Unit –IV	**09 Hrs**
Stars: Stellar Spectra, The Harvard Spectral Classification, The Hertzsprung–Russell Diagram, Binary Stars and Stellar Masses - Classification of Binary Stars, Stellar Structure - Internal Equilibrium Conditions, The Sun - Sun's Atmosphere, Solar Activity; Stellar Energy Sources - Nuclear Fusion Reactions, Stellar Evolution - Evolutionary Time Scales, The Contraction of Stars Towards the Main Sequence, The Main Sequence Phase, The Giant Phase, The Final Stages of Evolution - White Dwarfs, Neutron Stars, Black Holes.	
Unit –V	**08 Hrs**
Galaxies: The Classification of Galaxies, Masses, Galactic Structures, Structural Components of the Milky Way, Dynamics of Galaxies, Systems of Galaxies, The Origin and Evolution of Galaxies. The Interstellar Medium - Interstellar Dust, Interstellar Gas, Cosmic Rays. **Cosmology:** The Olbers Paradox, Hubble's Law, Cosmic Microwave Background Radiation, History of the Universe, Astrobiology.	

Course Outcomes	
CO1	Apply the principles of celestial mechanics to describe planetary motion, orbital dynamics, and gravitational interactions in astrophysical systems.
CO2	Compare and evaluate different types of telescopes and their applications in various regions of the electromagnetic spectrum for observational astronomy.
CO3	Interpret the foundational concepts of special relativity and analyze the implications of time dilation, length contraction, and relativistic mass in astronomical contexts.
CO4	Describe the structure and evolution of stars and galaxies and interpret key cosmological models to explain the origin, expansion, and fate of the universe.

Reference Books	
1.	Bradley W. Carroll, Dale A. Ostlie, An Introduction to Modern Astrophysics: United States Edition, Pearson; 1st edition (7 September 2017), ISBN-10: 1108422160
2.	H. Karttunen, P. Kröger, H. Oja, M. Poutanen, K. J. Donner (Eds.), Fundamental Astronomy, 6th Edition, Springer, 2017, ISBN 978-3662571033
3.	Baidyanath Basu, An Introduction to Astrophysics, 2nd Edition, Prentice Hall India, 2013, ISBN : 978-8120340718

4.	Suresh Chandra Mohit Kumar, A Textbook of Astronomy and Astrophysics, Dreamtech Press, 1 Nov 2019, ISBN-10: 9389520908
5.	Padmanabhan, T., Theoretical Astrophysics, Vols.1-3, 2015, Cambridge University Press, ISBN-9780521016278
6.	Modern Cosmology, Academic Press Inc; 2nd edition, 7 July 2020, ISBN-10: 0128159480,
7.	Harwit, M. Astrophysical Concepts, 4th Edition, 2015, Springer-Verlag, ISBN-10 : 1441921990
8.	Shapiro, Stuart L., and Saul A. Teukolsky. Black Holes, White Dwarfs, and Neutron Stars, 1st Edition, 1983, Wiley, ISBN: 9780471873167

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	**COMPONENTS**	**MARKS**
1.	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. **THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.**	**20**
2.	**TESTS:** Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). THREE tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 150 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) **ADDING UPTO 40 MARKS**.	**40**
	MAXIMUM MARKS FOR THE CIE (THEORY AND PRACTICE)	**100**

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	20
	PART B (Maximum of TWO Sub-divisions only)	
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
	TOTAL	**100**

Semester: VII						
MATHEMATICS OF MUSIC **Institutional Elective – II (Group G)** **(Theory)**						
Course Code	**:**	**MA375TGR**		**CIE**	**:**	**100 Marks**
Credits: L:T:P	**:**	**3:0:0**		**SEE**	**:**	**100 Marks**
Total Hours	**:**	**45L**		**SEE Duration**	**:**	**3 Hours**

Unit-I	**09 Hrs**
Fundamentals of Sound and Musical Structure Nature of sound: frequency, amplitude, harmonics, waveforms; Musical pitch and logarithmic perception of frequency, Shruti, Swara, Saptak; Musical notation systems.	
Unit – II	**09 Hrs**
Tuning Systems and Modular Arithmetic in Music Pythagorean tuning, Just Intonation, and Equal Temperament, Circle of Fifths and the irrationality of log(3/2), Modular arithmetic and pitch class sets, Construction of scales: Melakarta Rāga(Carnatic) and Thaat (Hindustani) systems.	
Unit –III	**09 Hrs**
Rhythm, Tāla, and Combinatorial Structures Tāla systems: concepts of mātra, vibhāga, āvartana; Analysis of Carnatic tālas (Adi, Rupaka, Jhampa, etc.); Combinatorics of tāla patterns and variation (korvais, tihāis); Algorithmic composition of rhythmic phrases	
Unit –IV	**09 Hrs**
Timbre, Music Analysis and Processing Timbre and harmonic spectra, Fourier series and transforms in sound decomposition, Sub-harmonics, Limitation of Fourier Analysis: Noise; Introduction to sound synthesis and spectral envelopes.	
Unit –V	**09 Hrs**
Algorithmic Music and Perception Fractals and recursion in music, Algorithmic and generative composition, Mathematical models of musical perception, Psychoacoustics: pitch, consonance/dissonance, critical bands.	

Course Outcomes	
CO1	Analyze musical phenomena using mathematical tools
CO2	Model rhythm, harmony, and timbre using algebraic and computational methods.
CO3	Apply signal processing concepts to sound synthesis and analysis.
CO4	Explore algorithmic and generative approaches to musical composition.

Reference Books	
1.	Benson, D. J. (2006). Music: A Mathematical Offering. Cambridge University Press, ISBN: 978-0521619998.
2.	Fauvel, J., Flood, R., & Wilson, R. (Eds.). (2003). Music and Mathematics: From Pythagoras to Fractals. Oxford University Press, ISBN: 978-0199298938.
3.	Loy, G. (2006). Musimathics: The Mathematical Foundations of Music (Vol. 1 & 2). MIT Press, ISBN: 9780262516556.
4.	Sethares, W. A. (2005). Tuning, Timbre, Spectrum, *Scale* (2nd ed.). Springer, ISBN: 978-1852337971.
5	Radhika Iyer (2018), Elements Of Indian Music: The Melakarta System, ISBN: 1513460854.

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	COMPONENTS	MARKS
1.	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. **THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.**	**20**
2.	**TESTS:** Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). **TWO tests will be conducted.** Each test will be evaluated for 50 Marks, adding upto 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20**) ADDING UPTO 40 MARKS**.	**40**
	MAXIMUM MARKS FOR THE CIE (THEORY AND PRACTICE)	**100**

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	20
	PART B (Maximum of TWO Sub-divisions only)	
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
	TOTAL	**100**

Semester: VII						
COGNITIVE PSYCHOLOGY **Institutional Elective Course (Group G)** **Theory**						
Course Code	**:**	**HS375TGS**		**CIE**	**:**	**100**
Credits: L:T:P	**:**	**03**		**SEE**	**:**	**100**
Total Hours	**:**	**45 Hrs**		**SEE Duration**	**:**	**3 Hours**

Unit-I	**09 Hrs**
Fundamentals & current trends in cognitive psychology: Definition, Emergence of cognitive psychology, Cognitive development theories and perspectives; Current status and trends in cognitive Psychology. Research methods in cognitive psychology- goals of research. Distinctive research method. Current areas of research in cognitive psychology, (Educational application, marketing and advertisement).	
Unit – II	**09 Hrs**
Basic cognitive processes: Sensation and Perception: Sensory receptors and Brain, The constancies, pattern recognition, Modularity, Imagery: Characteristics of Imagery, Cognitive maps. Attention and Information processing: Nature and Types, Theories and models of attention. Neuropsychological studies of Attention. Consciousness: – meaning, Modern Theories and Contemporary Research of Consciousness.	
Unit –III	**09 Hrs**
Reasoning, Creativity and Problem- Solving: Reasoning definition, types, influencing factors. Creativity-definition, steps involved in creative process, obstacles involved in creativity, enhancing techniques of creativity. Meta cognition: Problem solving, steps in problem solving, types, methods, obstacles and aids of problem Solving.	
Unit –IV	**09 Hrs**
Psycholinguistics: Definition, characteristics of language, theories - Chomsky. Structure of Language (Properties), Stages in Language Development, Neurological Language. Comprehension and Production. Bilingualism, Multilingualism and Learning disability.	
Unit –V	**09 Hrs**
Cognitive Neuroscience: Definition and emergence of cognitive neuroscience, Scope of Neuroscience, structure and functions of Brain, Brain Plasticity, Intelligence and Neuroscience.Meta-cognitive strategies. Artificial intelligence, Robotics, Models on Information Processing.	

Course Outcomes: After completing the course, the students will be able to: -	
CO1	Describe the basic theories, principles, and concepts of cognitive psychology as they relate to behaviours and mental processes.
CO2	Define learning and compare and contrast the factors that cognitive, behavioural, and Humanistic theorists believe influence the learning process.
CO3	Develop understanding of psychological attributes such as reasoning, problem solving creativity, resulting in their enhancement and apply effective strategies for self-management and self-improvement.
CO4	Apply the theories into their own and others' lives in order to better understand their personalities and experiences.

Reference Books	
1.	Sterberg R.J and Sternberg Karin(2012) Cognitive Psychology 6th Edition Woods worth Cenguage Learning
2.	Psychology-themes and variations , Wayne Weiten, IV edition, Brooks / Cole Publishing Co.
3.	Psychology Robert A. Baron, III edition (1995) Prentice Hall India.
4.	Understanding Psychology Feldman R. S, IV edition, (1996) McGraw Hill India

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	COMPONENTS	MARKS
1.	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. **THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.**	**20**
2.	**TESTS:** Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding up to 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar / presentation / demonstration (20**) ADDING UPTO 40 MARKS**.	**40**
MAXIMUM MARKS FOR THE CIE		**100**

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	20
	PART B (Maximum of TWO Sub-divisions only)	
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
	TOTAL	**100**

Semester: VII

PRINCIPLES & PRACTICES OF CYBER LAW
Institutional Elective Course (Group G)
Theory

Course Code	**:**	**HS375TGT**		**CIE**	**:**	**100 Marks**
Credits: L:T:P	**:**	**3:0:0**		**SEE**	**:**	**100 Marks**
Total Hours	**:**	**45L**		**SEE Duration**	**:**	**3 Hours**

Unit-I — **08 Hrs**

Introduction - Origin and meaning of Cyberspace; Introduction to Indian Cyber Law, Distinction between Cyber Crime and Conventional Crime,Cyber Criminals and their Objectives, Kinds of Cyber Crime& Cyber Threats, challenges of cybercrimes,Overview of General Laws and Procedures in India.
Cyber Jurisdiction-Concept of Jurisdiction, Jurisdiction in Cyberspace, Issues and concerns of Cyberspace Jurisdiction in India, International position of Cyberspace Jurisdiction, Judicial interpretation of Cyberspace Jurisdiction.
Activities:Case Studies and Practical Applications

Unit – II — **08 Hrs**

Information Technology Act: A brief overview of Information Technology Act 2000, IT Act 2000 vs. IT Amendment Act 2008, Relevant provisions from Indian Penal Code, Indian Evidence Act, Bankers Book Evidence Act, Reserve Bank of India Act, etc.
Electronic Signature and Digital Signature- Meaning & Concept of Relevance of Signature, Handwritten signature vs Digital Signature, Technological Advancement and development of signature, Digital Signature: IT Act, 2000, Cryptography, Public Key and Private Key, Public Key Infrastructure Electronic Signature vs. Digital Signature, E-Commerce under IT Act2000, Issues and challenges of E-Commerce.
Activities: Case Studies and Practical Applications

Unit –III — **08 Hrs**

Data Protection and Privacy Concerns in Cyberspace - Need to protect data in cyberspace, Types of data, Legal framework of data protection, Data protection bill -an overview, GDPR, Concept of privacy, Privacy concerns of cyberspace, Constitutional framework of privacy, Judicial interpretation of privacy in India.
Data Privacy and Data Security- Defining data, meta-data, big data, non- personal data. Data protection, Data privacy and data security, Data protection regulations of other countries- General Data Protection Regulations (GDPR),2016 Personal Information Protection and Electronic Documents Act (PIPEDA)., Social media- data privacy and security issues.
Activities: Case Studies and Practical Applications

Unit –IV — **08 Hrs**

IP Protection Issues in Cyberspace
Copyright Issues in Cyberspace- Copyright infringement in digital environment. Indian legal protection of copyright in cyberspace.
Trademark Issues in Cyberspace -Domain Name Vs Trademark,Domain Name dispute and Related Laws, Different Form of Domain in Cyberspace.
Patent Issues in Cyberspace-Legal position on Computer related Patents - Indian Position on Patents.
Activities:Case Studies and Practical Applications

Unit –V — **07 Hrs**

Digital Forensics- Computer Forensics, Mobile Forensics,Forensic Tools ,Anti-Forensics
Cyber Crime & Criminal Justice Agencies -Cyber Crime Cells, Cyber Crime Appellate- Cyber Crime Investigation, Investigation Procedure- FIR -Charge Sheet

Course Outcomes: After completing the course, the students will be able to: -	
CO1	Understand the importance of professional practice, Law and Ethics in their personallives and professional careers.
CO2	Build in Depth Knowledge of Information Technology Act and Legal Frame Work of Right to Privacy, Data Security and Data Protection.
CO3	Identify the bone of contentions of cybercrime investigation techniques, evaluate problem-solving strategies, and develop science-based solutions.
CO4	Develop an Understanding of the Relationship Between E-Commerce and Cyberspace.

Reference Books	
1.	Cyber Law by Dr. Pavan Duggal Publisher: LexisNexis, ISBN-10: 8196241070, ISBN-13: 978-8196241070
2.	Introduction to Information Security and Cyber Laws by Surya Prakash Tripathi, Ritendra Goel, Praveen Kumar Shukla ASIN: 9351194736,Publisher: Dreamtech Press, ISBN-10: 9789351194736,ISBN-13: 978-9351194736.
3.	Cyber Forensics in India: A Legal Perspective by Nishesh Sharma,1st Edition, ISBN: 9788131250709.
4.	Cyber Laws,Justice Yatindra Singh, 6th Edition, Vol. 1, ISBN : 9789351437338

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	**COMPONENTS**	**MARKS**
1.	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. **THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.**	**20**
2.	**TESTS:** Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding up to 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar / presentation / demonstration (20) **ADDING UPTO 40 MARKS**.	**40**
MAXIMUM MARKS FOR THE CIE		**100**

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	20
	PART B (Maximum of TWO Sub-divisions only)	
2	Unit 1 : (Compulsory)	16
3 & 4	Unit 2 : Question 3 or 4	16
5 & 6	Unit 3 : Question 5 or 6	16
7 & 8	Unit 4 : Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
	TOTAL	**100**

Semester: VII						
SUMMER INTERNSHIP						
Course Code	**:**	**IM376SI**		**CIE**	**:**	**100 Marks**
Credits: L:T:P	**:**	**0:0:3**		**SEE**	**:**	**100 Marks**
No. of Weeks	**:**	**08**		**SEE Duration**	**:**	**03 Hours**

GUIDELINES

1. The duration of the internship shall be for a period of 8 weeks on full time basis after VI semester final exams and before the commencement of VII semester.
2. The student must submit letters from the industry clearly specifying his / her name and the duration of the internship on the company letter head with authorized signature.
3. Internship must be related to the field of specialization of the respective UG programme in which the student has enrolled.
4. Students undergoing internship training are advised to report their progress and submit periodic progress reports to their respective guides.
5. Students have to present the internship activities carried out to the departmental committee and only upon approval by the committee, the student can proceed to prepare and submit the hard copy of the final internship report. However, interim or periodic reports as required by the industry / organization can be submitted as per the format acceptable to the respective industry /organizations.
6. The reports shall be printed on A4 size with 1.5 spacing and Times New Roman with font size 12, outer cover of the report (wrapper) has to be Ivory color for UG circuit Programs and Light Blue for Non-Circuit Programs.
7. The broad format of the internship final report shall be as follows
    - Cover Page
    - Certificate from College
    - Certificate from Industry / Organization
    - Acknowledgement
    - Synopsis
    - Table of Contents
    - Chapter 1 - Profile of the Organization: Organizational structure, Products, Services, Business Partners, Financials, Manpower, Societal Concerns, Professional Practices,
    - Chapter 2 - Activities of the Department
    - Chapter 3 - Tasks Performed: summary of the tasks performed during 8-week period
    - Chapter 4 – Reflections: Highlight specific technical and soft skills acquired during internship
    - References & Annexure

Course Outcomes:

After going through this internship, the student shall be able to:

CO1: Apply Engineering and Management principles

CO2: Analyze real-time problems and suggest alternate solutions

CO3: Communicate effectively and work in teams

CO4: Imbibe the practice of professional ethics and need for lifelong learning.

Scheme of Continuous Internal Evaluation (CIE):

The evaluation committee shall consist of Guide, Professor/Associate Professor and Assistant Professor. The committee shall assess the presentation and the progress reports in two reviews. The evaluation criteria shall be as per the rubrics given below:

Reviews	Activity	Weightage
Review-I	Explanation of the application of engineering knowledge in industries, ability to comprehend the functioning of the organization/ departments.	50 Marks
Review - II	Importance of resource management, environment and sustainability, presentation skills and report writing	50 Marks

Scheme for Semester End Evaluation (SEE):
The SEE examination shall be conducted by an external examiner (domain expert) and an internal examiner. Evaluation shall be done in batches, not exceeding 6 students per batch.

Scheme of Evaluation for SEE	
Particulars	**%Marks**
Project Synopsis (Initial Write-up)	10%
Project Demo/Presentation	30%
Methodology and Results Discussion	30%
Project Work Report	10%
Viva-voce	20%
Total	**100**

Semester: VIII						
MAJOR PROJECT						
Course Code	**:**	**IM481P**		**CIE**	**:**	**100 Marks**
Credits: L:T:P	**:**	**0:0:12**		**SEE**	**:**	**100 Marks**
Hours/Week	**:**	**24**		**SEE Duration**	**:**	**03 Hours**

GUIDELINES

1. The project topic, title and synopsis have to be finalized and submitted to their respective internal guide(s) before the beginning of the 8th semester.
2. The detailed Synopsis (approved by the department Project Review Committee) has to be submitted during the 1st week after the commencement of 8th semester.

Batch Formation:

- Students are free to choose their project partners from within the program or any other program.
- Each student in the team must contribute towards the successful completion of the project.
- The project may be carried out In-house / Industry / R & D Institution. The project work is to be carried out by a team of two to four students, in exceptional cases where a student is placed in a company and offered an internship through the competitive process or student is selected for internship at national or international level through competitive process, the student can work independently.
- The students are allowed to do either a project for full 5 days in the industry or full 5 days in the college.
- In case the project work is carried out outside Bengaluru, such students must be available during Project Evaluation process scheduled by the respective departments and they must also interact with their guide regularly through Email / Webinar / Skype etc.

Project Topic Selection:

The topics of the project work must be in the field of respective program areas or in line with CoE's(Centre of Excellence) identified by the college or List of project areas as given by industry/Faculty. The projects as far as possible should have societal relevance with focus on sustainability.

Project Evaluation:

- Continuous monitoring of project work will be carried out and cumulative evaluation will be done.
- The students are required to meet their internal guides once in a week to report their progress in project work.
- Weekly Activity Report (WAR) has to be maintained in the form of a diary by the project batch and the same has to be discussed with the Internal Guide regularly.
- In case of Industry project, during the course of project work, the internal guides will have continuous interaction with external guides and will visit the industry at least twice during the project period.
- For CIE assessment the project groups must give a final seminar with the draft copy of the project report.
- The presentation by each group will be for 20-30 minutes and every member of the team needs to justify the contributions to the project.
- The project team is required to submit Hard copies of the detailed Project Report in the prescribed format to the department.
- For CIE 50% weightage should be given to the project guide and 50% weightage to the project evaluation committee.
- Before the final evaluations the project group is required to produce a No dues certificate from Industry, Central Library and Department.

Course Outcomes:

After going through this major project, the student will be able to:

CO1: Apply knowledge of mathematics, science and engineering to solve respective engineering domain problems.

CO2: Design, develop, present and document innovative/multidisciplinary modules for a complete engineering system.

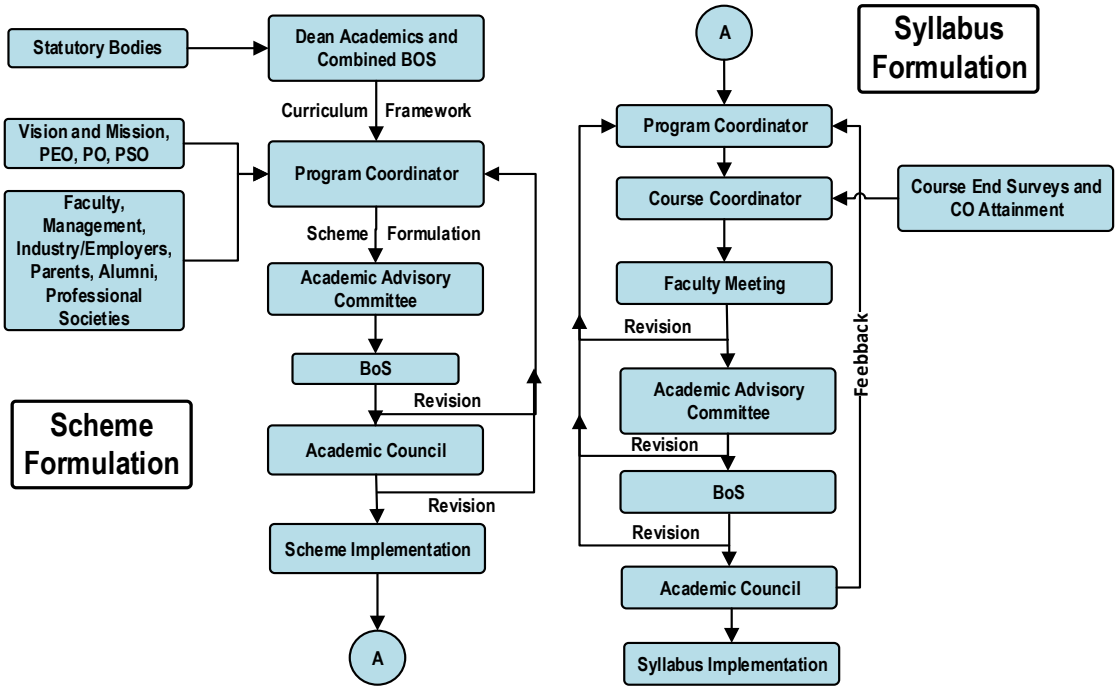
CO3: Use modern engineering tools, software and equipment to solve problem and engage in life-long learning to follow technological developments.

CO4: Function effectively as an individual, or leader in diverse teams, with the understanding of professional ethics and responsibilities.

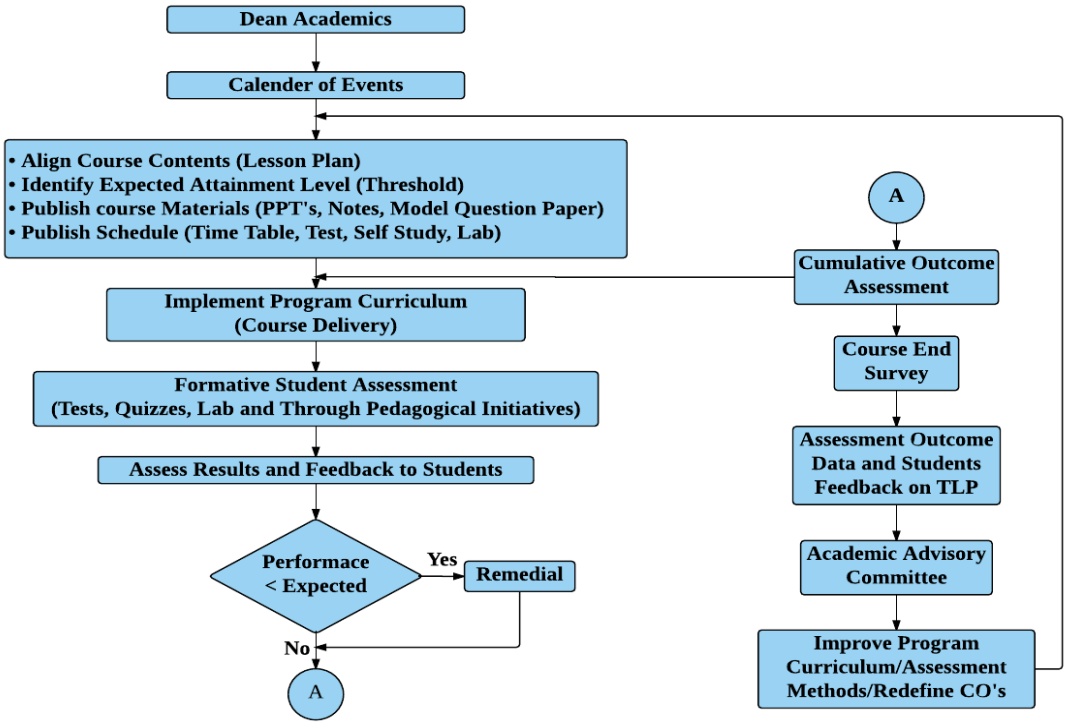
Scheme of Continuous Internal Evaluation (CIE):

Scheme of Evaluation for CIE		**Scheme of Evaluation for SEE**	
Particulars	**%Marks**	**Particulars**	**%Marks**
Project Evaluation I	10%	Project Synopsis (Initial Writeup)	10%
Project Evaluation II	20%	Project Demo/Presentation	25%
Project Evaluation III	25%	Methodology, Results and Discussion	25%
Project Evaluation Phase-IV (Submission of Draft Project Report for Verification) (Project Final Internal Evaluation)	25%	Project Work Report	15%
Publication	20%	Viva-voce	25%
Total	**100**	**Total**	**100**

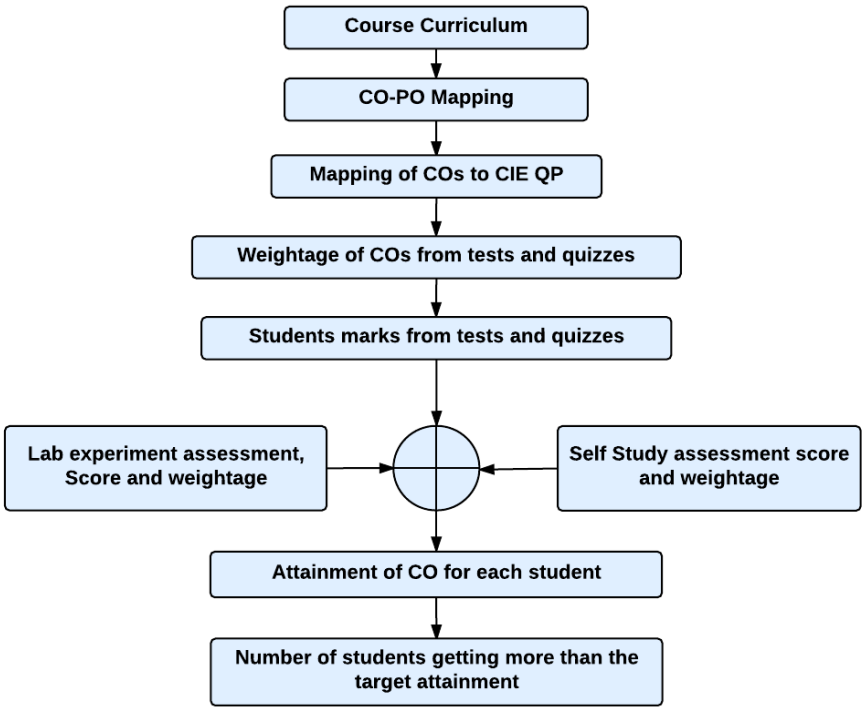
Curriculum Design Process



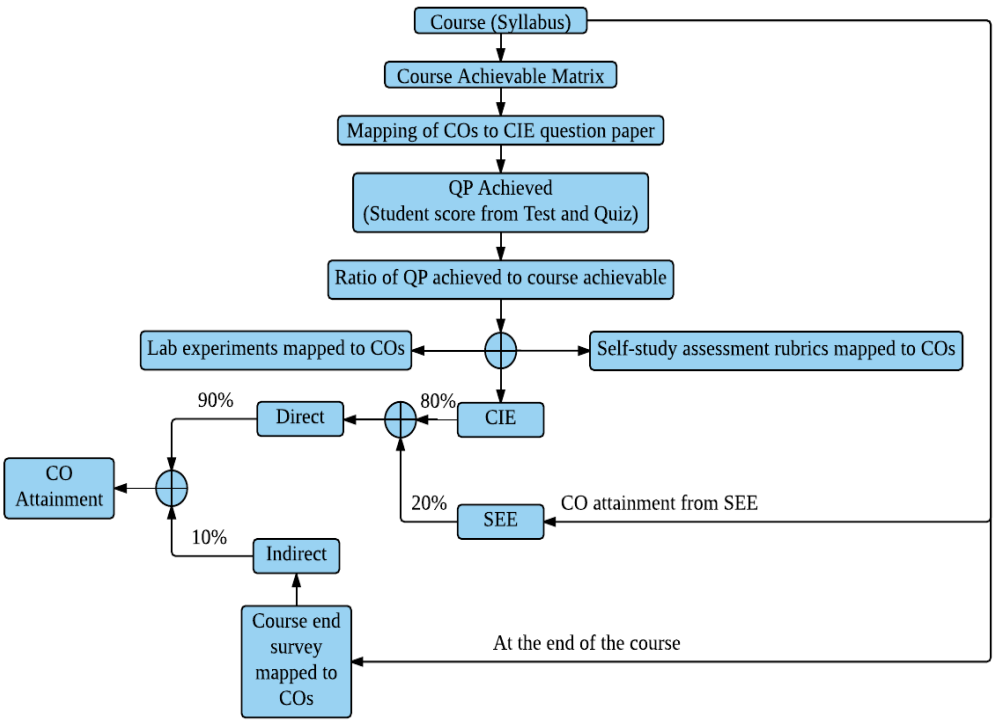
Academic Planning and Implementation



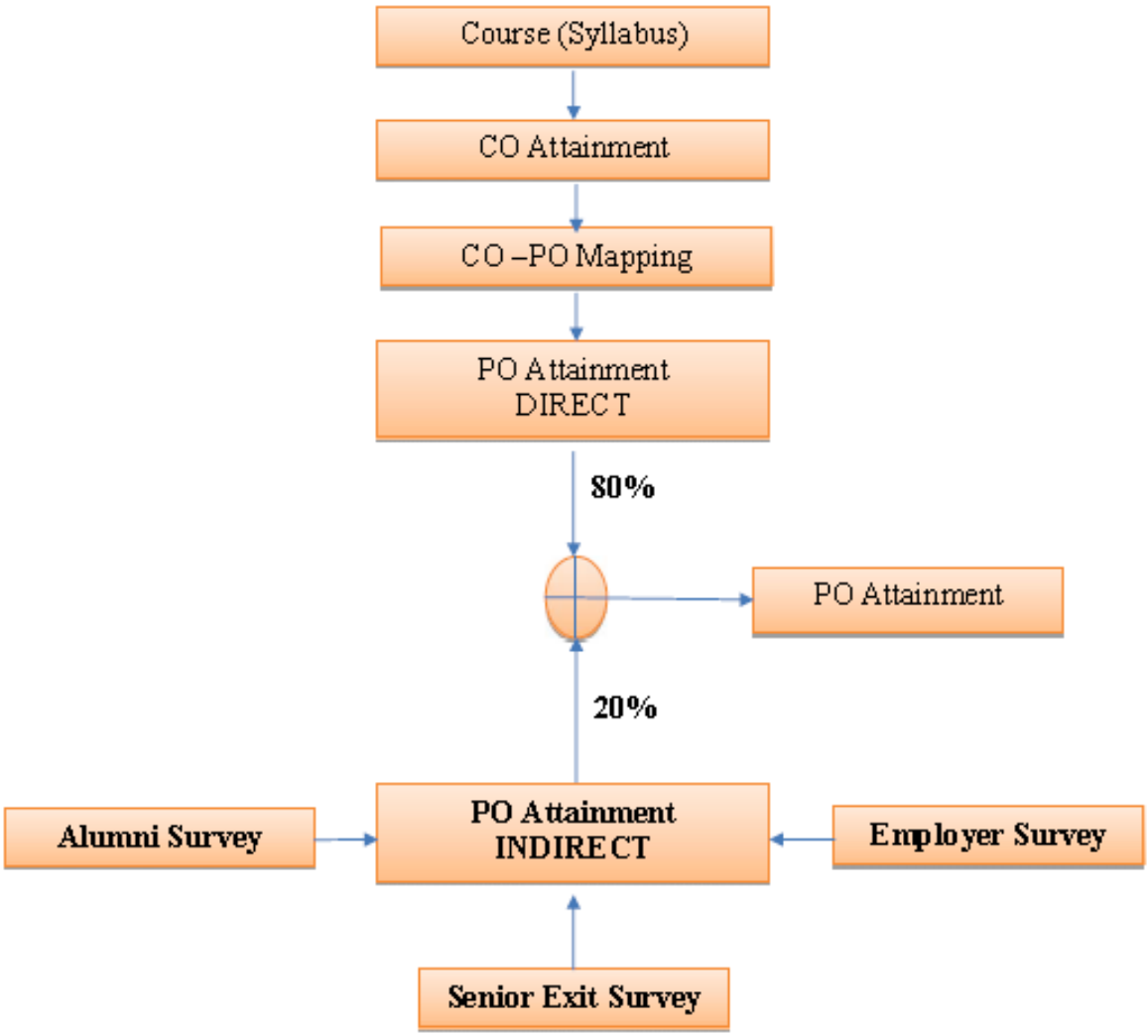
Process For Course Outcome Attainment



Final CO Attainment Process



Program Outcome Attainment Process



Knowledge and Attitude Profile (WK)

- **WK1:** A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.
- **WK2:** Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
- **WK3:** A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
- **WK4:** Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.
- **WK5:** Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.
- **WK6:** Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.
- **WK7:** Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.
- **WK8:** Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.
- **WK9:** Ethics, inclusive behaviour and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.

New Program Outcomes(PO)

- **PO1:** Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
- **PO2:** Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
- **PO3:** Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
- **PO4:** Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
- **PO5:** Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
- **PO6**: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
- **PO7:** Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
- **PO8:** Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
- **PO9:** Communication: Communicate effectively and inclusively within the community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences
- **PO10:** Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
- **PO11:** Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

INNOVATIVE TEAMS OF RVCE

Ashwa Mobility Foundation (AMF): Designs and fabricates Formula-themed race cars and mobility solutions to address urban transportation issues.

Astra Robotics Team: Focuses on designing and building application-specific robots.

Coding Club: Helps students gain coding skills and succeed in competitions like GSoC and ACM-ICPC.

Frequency Club Team: Works on software and hardware, emphasizing AI and Machine Learning.

Team Garuda: Develops a supermileage urban concept electric car and E-mobility products.

Team Jatayu: Builds low-cost UAVs with autonomous capabilities for various tasks.

Team Antariksh: Focuses on space technology and the development of operational rockets.

Team Chimera: Builds a Formula Electric Car through R&D in E-Mobility.

Helios Racing Team: Designs and tests All-Terrain Vehicles, participating in SAE's BAJA competitions.

Team Krushi: Creates low-cost farming equipment to assist farmers in cultivation and harvesting.

Team Vyoma: Designs and tests radio-controlled aircraft and UAVs.

Team Dhruva: Engages in astronomy-related activities and collaborates on projects with organizations like ICTS and IIA.

Ham Club: Promotes Amateur Radio and explores technical innovations in communications, especially for disaster response.

Chitrak : It is the official electric bike team of RV College of Engineering (RVCE) in Bangalore

Anoraniya:A Quantum based technical club initiated by students. The club is dedicated to addressing various technical challenges in quantum technologies and is on the verge of delivering some fascinating results.

Accelerate Club: organizes and participates in hackathons, bootcamps, workshops, and student-led innovation drives.

Cultural Activity Teams

1. AALAP (Music club)
2. DEBSOC (Debating society)
3. CARV (Dramatics club)
4. FOOTPRINTS (Dance club)
5. QUIZCORP (Quizzing society)
6. ROTARACT (Social welfare club)
7. RAAG (Youth club)
8. EVOKE (Fashion team)
9. f/6.3 (Photography club)
10. CARV ACCESS (Film-making

NSS of RVCE

NCC of RVCE



VISION

Leadership in Quality Technical Education, Interdisciplinary Research & Innovation, with a Focus on Sustainable and Inclusive Technology



MISSION

- To deliver outcome based Quality education, emphasizing on experientiallearning with the state of the art infrastructure.
- To create a conducive environment for interdisciplinary research and innovation.
- To develop professionals through holistic education focusing on individual growth, discipline, integrity, ethics and social sensitivity.
- To nurture industry-institution collaboration leading to competency enhancement and entrepreneurship.
- To focus on technologies that are sustainable and inclusive, benefiting all sections of the society.



QUALITY POLICY

Achieving Excellence in Technical Education, Research and Consulting through an Outcome Based Curriculum focusing on Continuous Improvement and Innovation by Benchmarking against the global Best Practices.



CORE VALUES

Professionalism, Commitment, Integrity, Team Work, Innovation